

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]
```

```

cols = list(dict.fromkeys(target_col + features_X +
                          feats_to_balance + feats_grouping +
                          feats_reference+ feats_postproc))

df = df[cols]
#df['campaign'] = "ALL"

return df

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r") as f:
            #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r") as f:
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")

    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)

```

```
#Removing of rabbit 4
df["column"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)

return df,dict_features
```

```
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Function print :

1.2 Metadata

Nombre de lignes df

7876

Nombre de lignes df clean	7876
Nombre de features	53
% de NaN	0.0
Taille dataset entraînement	5513
Taille dataset test	2362
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_2h_mean, r_NH3_st_2h_std, r_NH3_st_2h_rms, r_NH3_st_2h_median, r_NH3_st_2h_min, r_NH3_st_2h_max, r_NH3_st_2h_diff_last, r_NH3_st_2h_slope, r_NH3_st_8h_mean, r_NH3_st_8h_std, r_NH3_st_8h_rms, r_NH3_st_8h_median, r_NH3_st_8h_min, r_NH3_st_8h_max, r_NH3_st_8h_diff_last, r_NH3_st_8h_slope, r_temperature_st_2h_mean, r_temperature_st_2h_std, r_temperature_st_2h_rms, r_temperature_st_2h_median, r_temperature_st_2h_min, r_temperature_st_2h_max, r_temperature_st_2h_diff_last, r_temperature_st_2h_slope, r_temperature_st_8h_mean, r_temperature_st_8h_std, r_temperature_st_8h_rms, r_temperature_st_8h_median, r_temperature_st_8h_min, r_temperature_st_8h_max, r_temperature_st_8h_diff_last, r_temperature_st_8h_slope, r_umidity_st_2h_mean, r_umidity_st_2h_std, r_umidity_st_2h_rms, r_umidity_st_2h_median, r_umidity_st_2h_min, r_umidity_st_2h_max, r_umidity_st_2h_diff_last, r_umidity_st_2h_slope, r_umidity_st_8h_mean, r_umidity_st_8h_std, r_umidity_st_8h_rms, r_umidity_st_8h_median, r_umidity_st_8h_min, r_umidity_st_8h_max, r_umidity_st_8h_diff_last, r_umidity_st_8h_slope

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermat and cluster list

The following section has been done with a threshold choosen of : 0.8

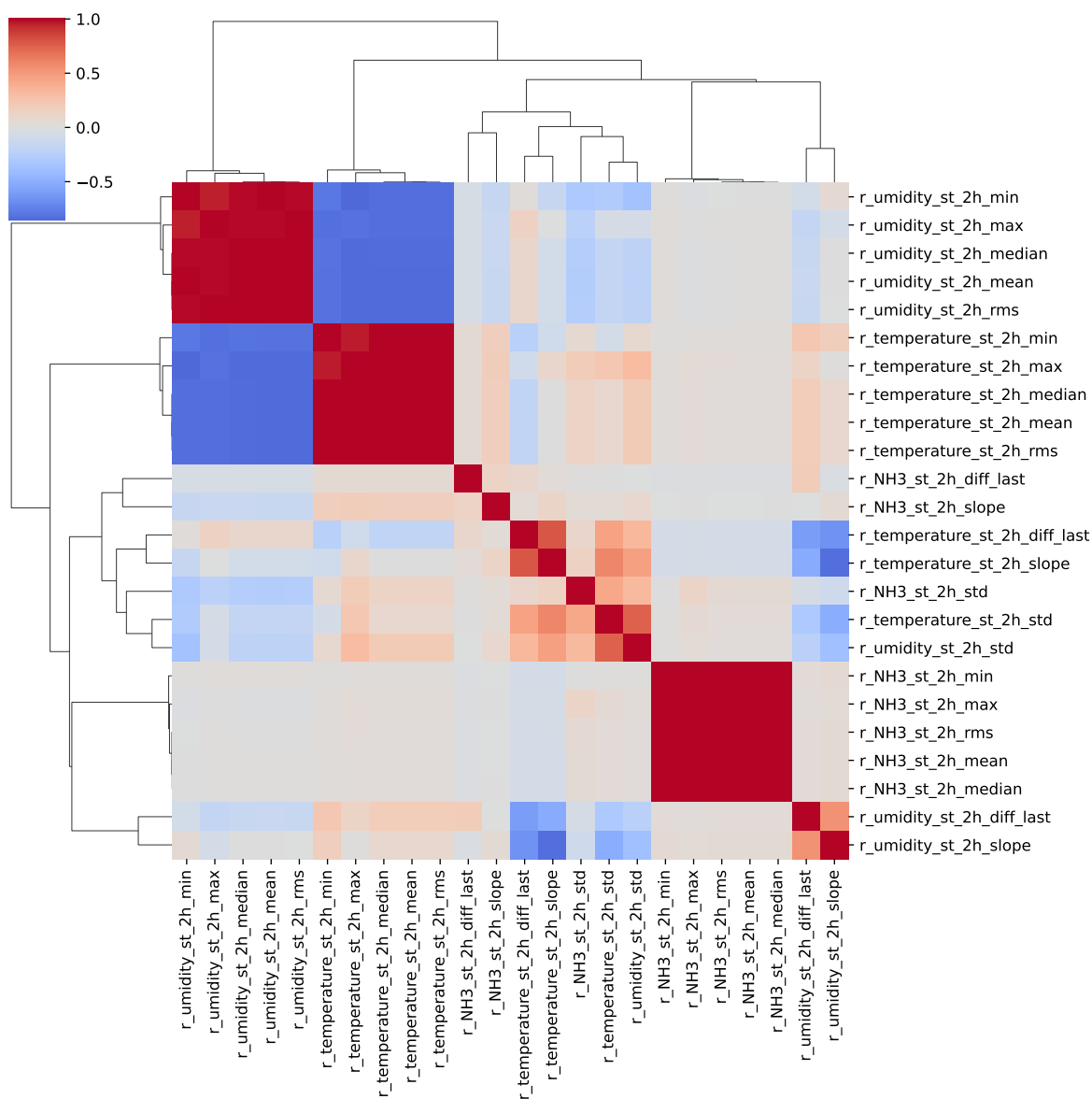
cluster 1: r, , N, H, 3 ; **cluster 2:** r, , t, e, m, p, e, r, a, t, u, r, e ; **cluster 3:** r, , u, m, i, d, i, t, y ; **cluster 4:** r, , C, O, 2 ; **cluster 5:** r, __, T, H, I ; **cluster 6:** r_NH3_st_2h_max,

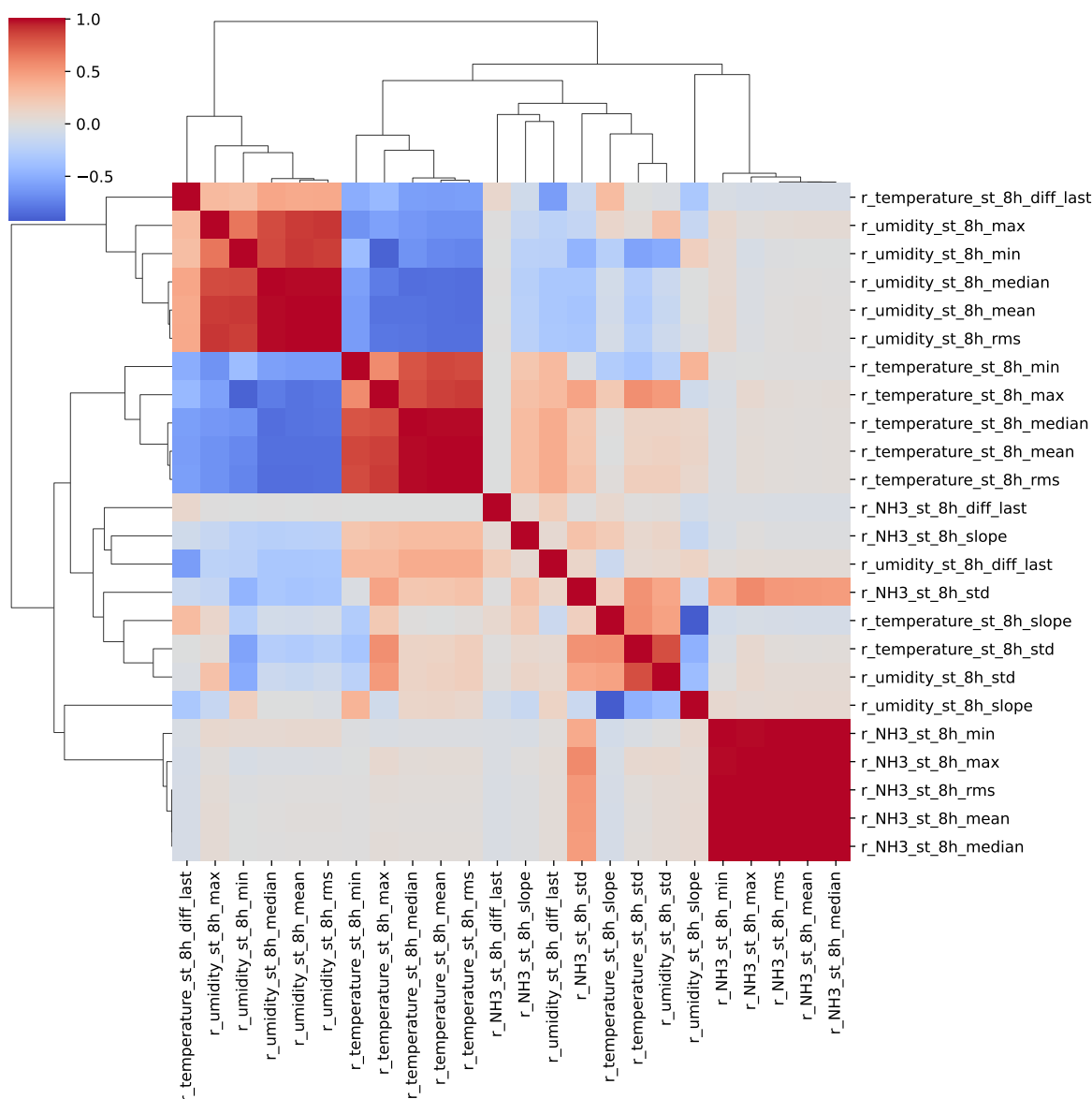
`r_NH3_st_2h_min`, `r_NH3_st_2h_rms`, `r_NH3_st_2h_median`, `r_NH3_st_2h_mean` ; **cluster 7:** `r_NH3_st_2h_std` ; **cluster 8:** `r_NH3_st_2h_diff_last` ; **cluster 9:** `r_NH3_st_2h_slope` ; **cluster 10:** `r_temperature_st_2h_mean`, `r_umidity_st_2h_max`, `r_temperature_st_2h_min`, `r_umidity_st_2h_rms`, `r_temperature_st_2h_max`, `r_temperature_st_2h_median`, `r_umidity_st_2h_mean`, `r_temperature_st_2h_rms`, `r_umidity_st_2h_min`, `r_umidity_st_2h_median` ; **cluster 11:** `r_temperature_st_2h_std` ; **cluster 12:** `r_temperature_st_2h_diff_last` ; **cluster 13:** `r_umidity_st_2h_slope`, `r_temperature_st_2h_slope` ; **cluster 14:** `r_umidity_st_2h_std` ; **cluster 15:** `r_umidity_st_2h_diff_last` ; **cluster 16:** `r_NH3_st_8h_max`, `r_NH3_st_8h_rms`, `r_NH3_st_8h_median`, `r_NH3_st_8h_min`, `r_NH3_st_8h_mean` ; **cluster 17:** `r_NH3_st_8h_std` ; **cluster 18:** `r_NH3_st_8h_diff_last` ; **cluster 19:** `r_NH3_st_8h_slope` ; **cluster 20:** `r_temperature_st_8h_min`, `r_temperature_st_8h_max`, `r_temperature_st_8h_mean`, `r_umidity_st_8h_min`, `r_umidity_st_8h_mean`, `r_umidity_st_8h_median`, `r_temperature_st_8h_rms`, `r_temperature_st_8h_median`, `r_umidity_st_8h_max`, `r_umidity_st_8h_rms` ; **cluster 21:** `r_temperature_st_8h_std`, `r_umidity_st_8h_std` ; **cluster 22:** `r_temperature_st_8h_diff_last` ; **cluster 23:** `r_temperature_st_8h_slope`, `r_umidity_st_8h_slope` ; **cluster 24:** `r_umidity_st_8h_diff_last` ;

2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Choosen features X : `r_NH3`, `r_temperature`, `r_umidity`, `r_CO2`, `r_THI`, `r_NH3_st_2h_max`, `r_NH3_st_2h_std`, `r_NH3_st_2h_diff_last`, `r_NH3_st_2h_slope`, `r_temperature_st_2h_mean`, `r_temperature_st_2h_std`, `r_temperature_st_2h_diff_last`, `r_umidity_st_2h_slope`, `r_umidity_st_2h_std`, `r_umidity_st_2h_diff_last`, `r_NH3_st_8h_max`, `r_NH3_st_8h_std`, `r_NH3_st_8h_diff_last`, `r_NH3_st_8h_slope`, `r_temperature_st_8h_min`, `r_temperature_st_8h_std`, `r_temperature_st_8h_diff_last`, `r_temperature_st_8h_slope`, `r_umidity_st_8h_diff_last`





3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_2h_max, r_NH3_st_2h_std, r_NH3_st_2h_diff_last, r_NH3_st_2h_slope, r_temperature_st_2h_mean, r_temperature_st_2h_std, r_temperature_st_2h_diff_last, r_umid-

ity_st_2h_slope, r_umidity_st_2h_std, r_umidity_st_2h_diff_last, r_NH3_st_8h_max, r_NH3_st_8h_std, r_NH3_st_8h_diff_last, r_NH3_st_8h_slope, r_temperature_st_8h_min, r_temperature_st_8h_std, r_temperature_st_8h_diff_last, r_temperature_st_8h_slope, r_umidity_st_8h_diff_last

Len(Features_X): 24

Config Gridsearch:{‘show’: True, ‘search_type’: ‘optuna’, ‘model_name’: None, ‘model’: None, ‘param_grid’: None, ‘best_param’: None, ‘cv_func’: ShuffleSplit(n_splits=5, random_state=42, test_size=0.3, train_size=None), ‘save_models’: True, ‘sumup-table’: True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_{\text{test}} < 0.95$ (under performance of the test)

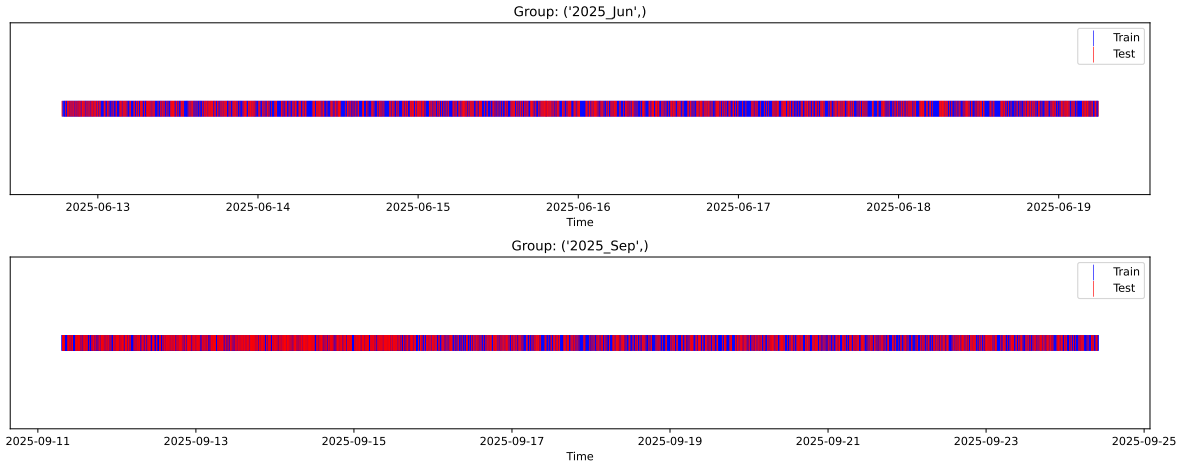
[Green] $r_{\text{test}} > 1.05$ (over performance of the test)

[Red] r_{train} not in $[0.95, 1.05]$ (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_train vs X_test

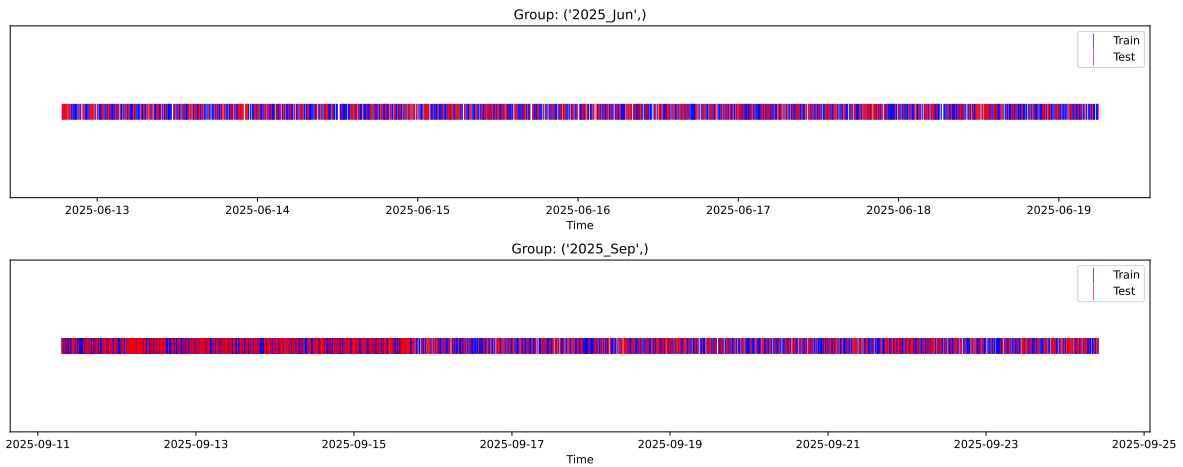
External Split: Train: 5513 samples Test: 2363 samples



3.3.2 Internal CV Folds

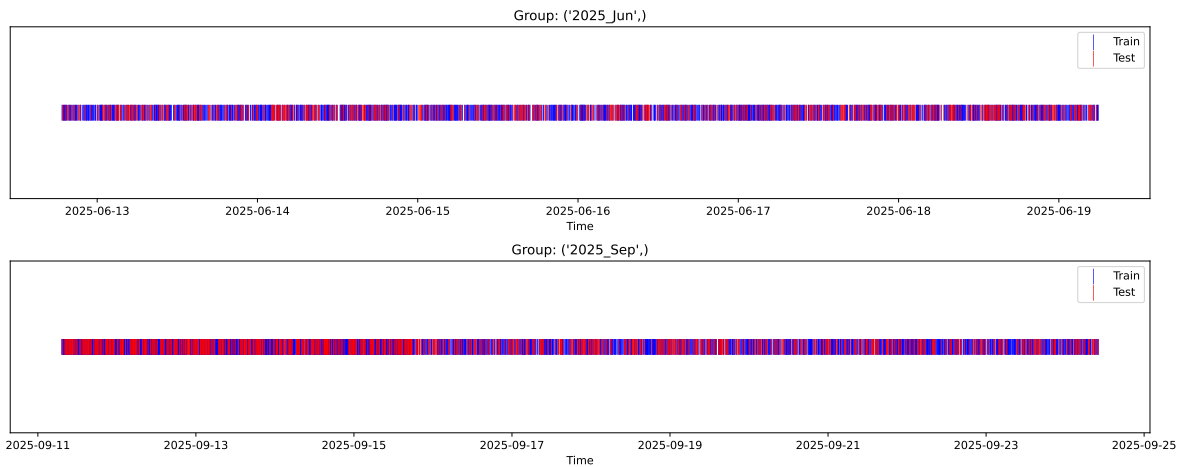
Fold 1/5

Train: 3859 samples Test: 1654 samples Excluded: 0 samples Window exclusion: 0h



Fold 2/5

Train: 3859 samples Test: 1654 samples Excluded: 0 samples Window exclusion: 0h



Fold 3/5

Train: 3859 samples Test: 1654 samples Excluded: 0 samples Window exclusion: 0h



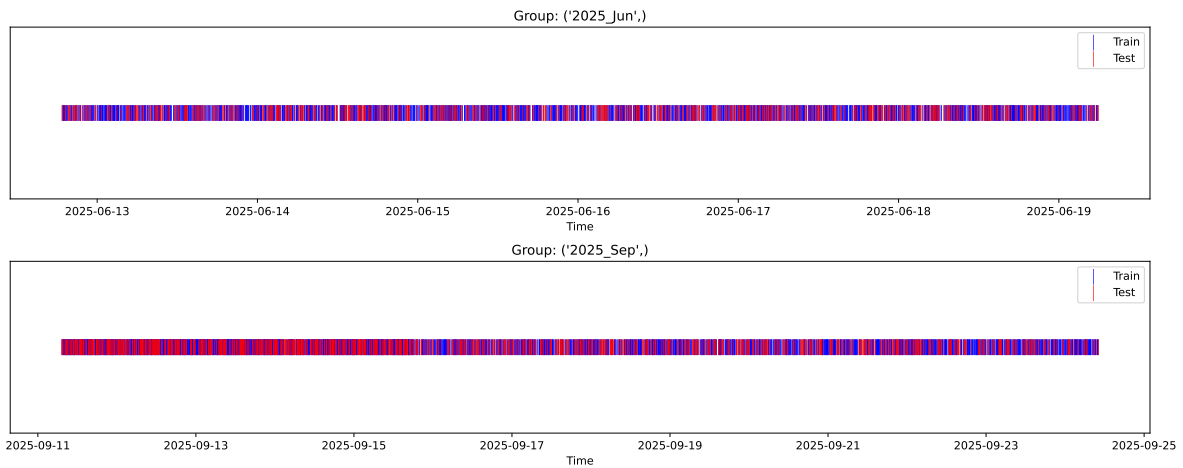
Fold 4/5

Train: 3859 samples Test: 1654 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3859 samples Test: 1654 samples Excluded: 0 samples Window exclusion: 0h



3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.354, std=0.811, min=0.959, max=13.845
 y_{test} : mean=2.358, std=0.777, min=1.068, max=7.822

Fold 0 — y_{test} : mean=2.35, std=0.80, y_{train} : mean=2.35, std=0.82

Fold 1 — y_{test} : mean=2.32, std=0.78, y_{train} : mean=2.37, std=0.82

Fold 2 — y_{test} : mean=2.35, std=0.85, y_{train} : mean=2.35, std=0.79

Fold 3 — y_test: mean=2.36, std=0.78, y_train: mean=2.35, std=0.83

Fold 4 — y_test: mean=2.32, std=0.75, y_train: mean=2.37, std=0.84

train scores CV: [0.53915001 0.54262323 0.55905908 0.54581554 0.54255248]

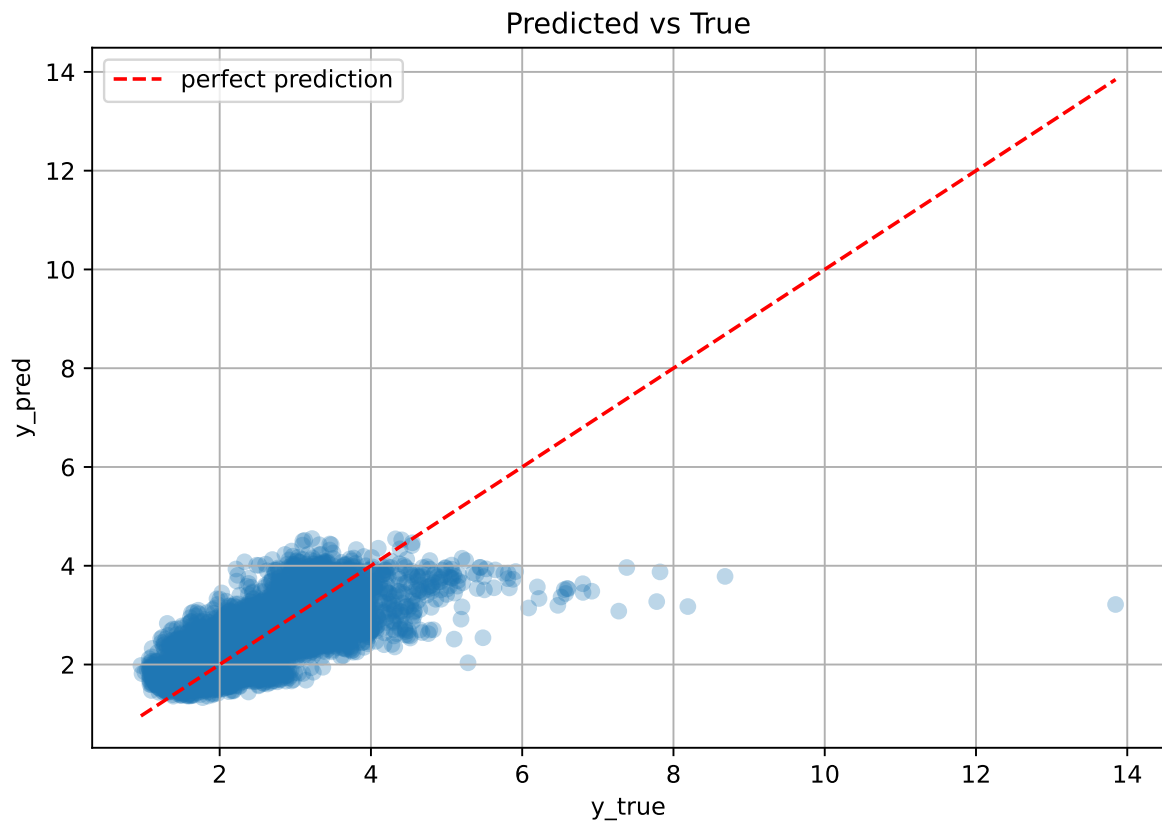
test scores CV: [0.55965293 0.55004184 0.51591487 0.54378717 0.54989407]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.5458	0.5439	0.0149	0.5784	0.5796	0.0357	1.0657	1.0036

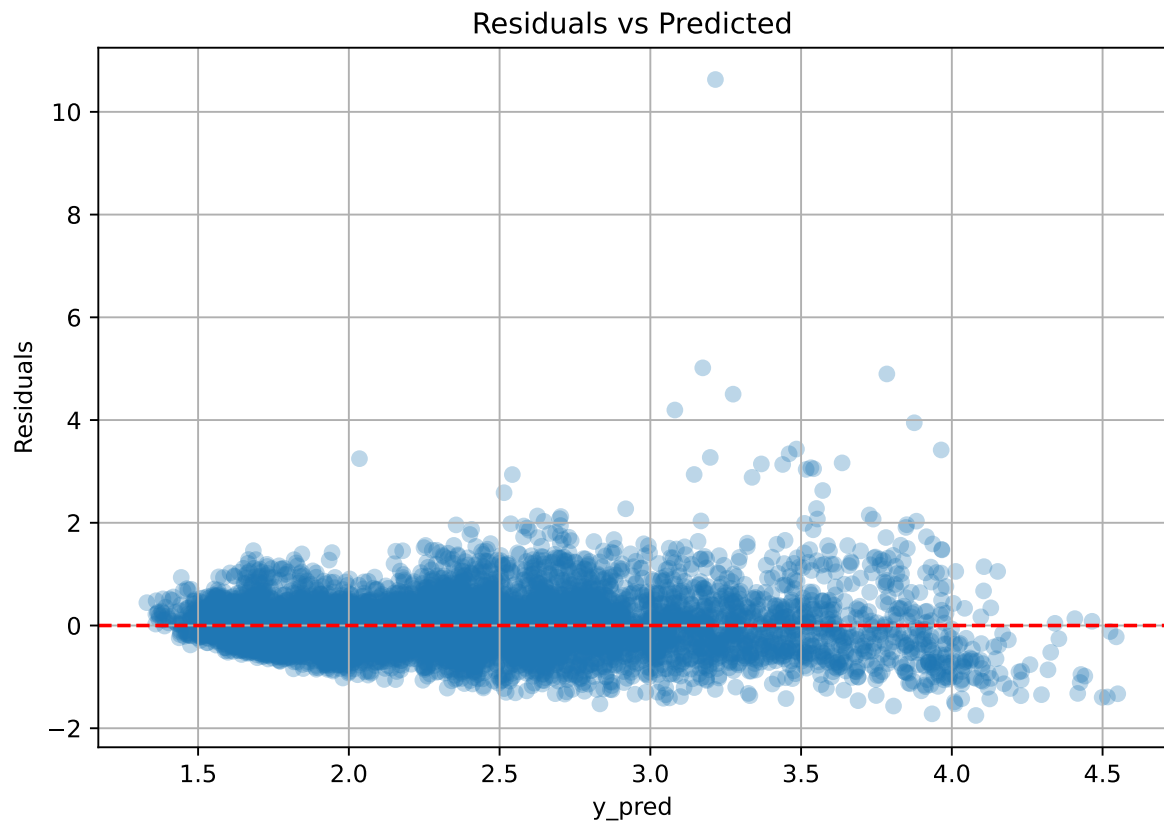
Coefficients:

coef_r_CO2 = -0.0433, coef_r_NH3 = -0.0716, coef_r_NH3_st_2h_diff_last = 0.0094, coef_r_NH3_st_2h_max = -0.3155, coef_r_NH3_st_2h_slope = -0.0113, coef_r_NH3_st_2h_std = 0.0643, coef_r_NH3_st_8h_diff_last = 0.0094, coef_r_NH3_st_8h_max = 0.3282, coef_r_NH3_st_8h_slope = 0.1040, coef_r_NH3_st_8h_std = -0.0057, coef_r_THI = 0.2361, coef_r_temperature = 0.6800, coef_r_temperature_st_2h_diff_last = -0.0126, coef_r_temperature_st_2h_mean = -0.8947, coef_r_temperature_st_2h_std = 0.0792, coef_r_temperature_st_8h_diff_last = -0.0126, coef_r_temperature_st_8h_min = 0.0156, coef_r_temperature_st_8h_slope = -0.0207, coef_r_temperature_st_8h_std = -0.0194, coef_r_umidity = -0.3662, coef_r_umidity_st_2h_diff_last = -0.0162, coef_r_umidity_st_2h_slope = -0.0363, coef_r_umidity_st_2h_std = 0.0479, coef_r_umidity_st_8h_diff_last = -0.0162, intercept = 2.3540,

3.4.2 Scatter



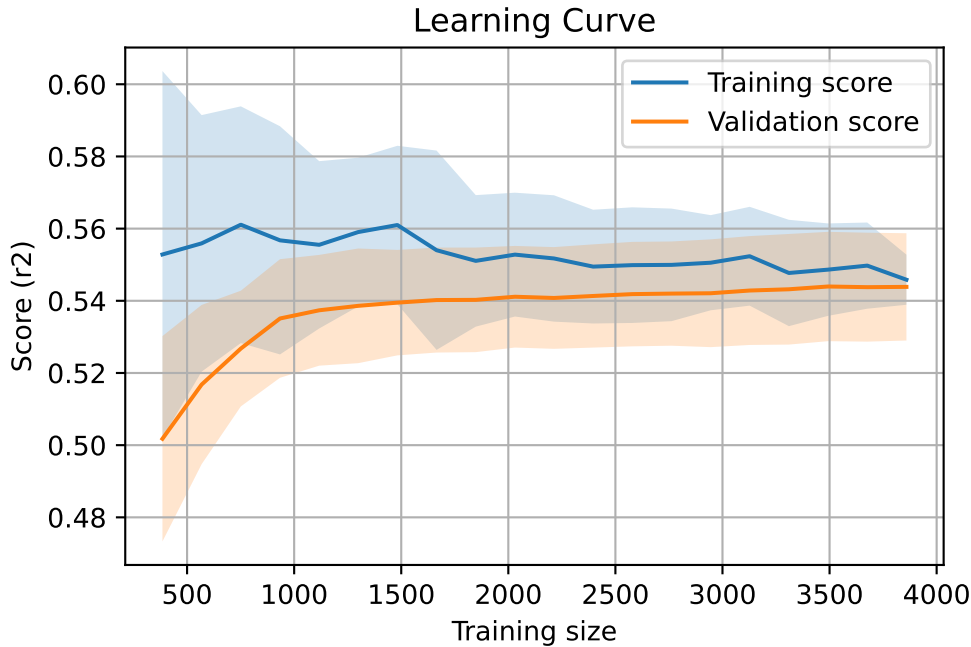
3.4.3 Scatter residuals



3.4.4 Learning curve

`Len(group)` : 7876, `Len(training)` : 5514,

`Len(training_real)` : 3860, `Len(test_of_training)` : 1654, `Len(test)` : 2362



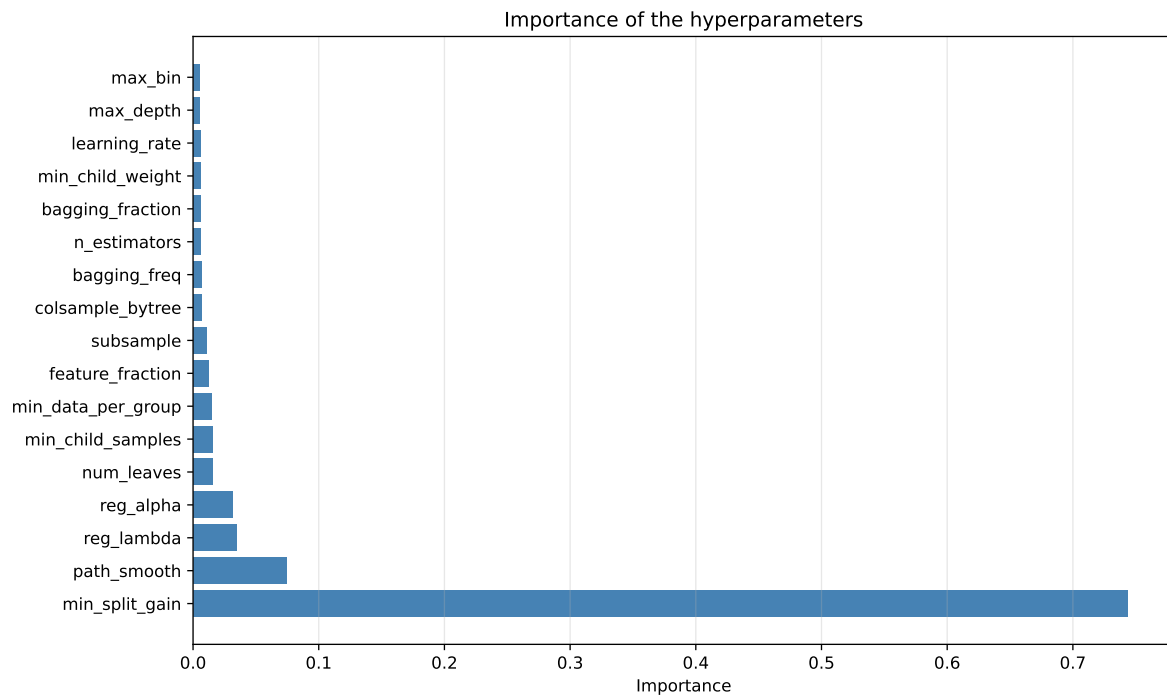
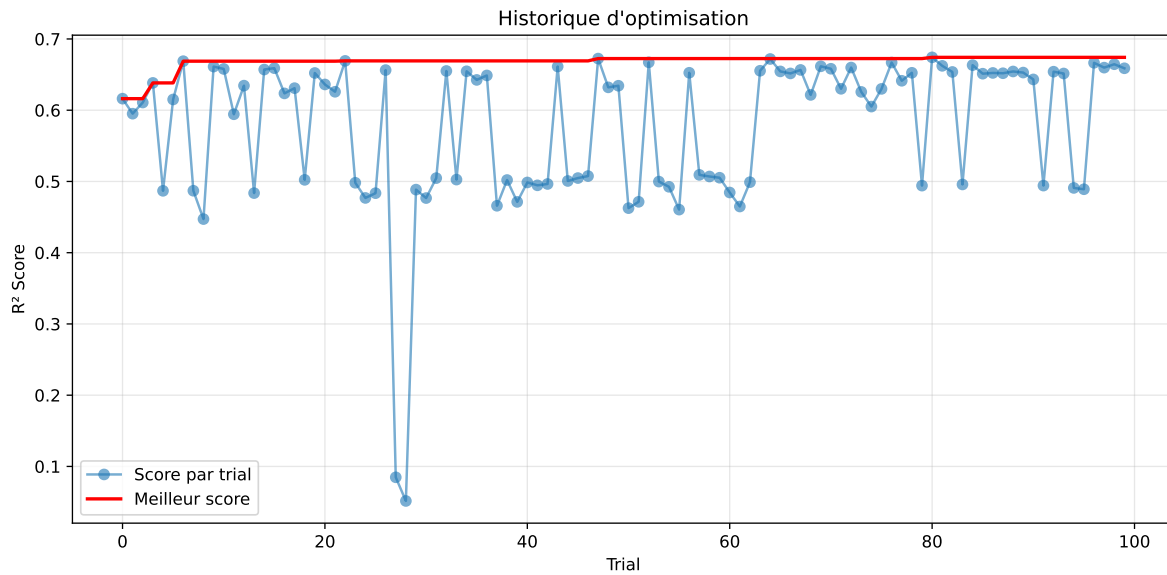
3.5 Model : LGBM

3.5.1 Gridsearch

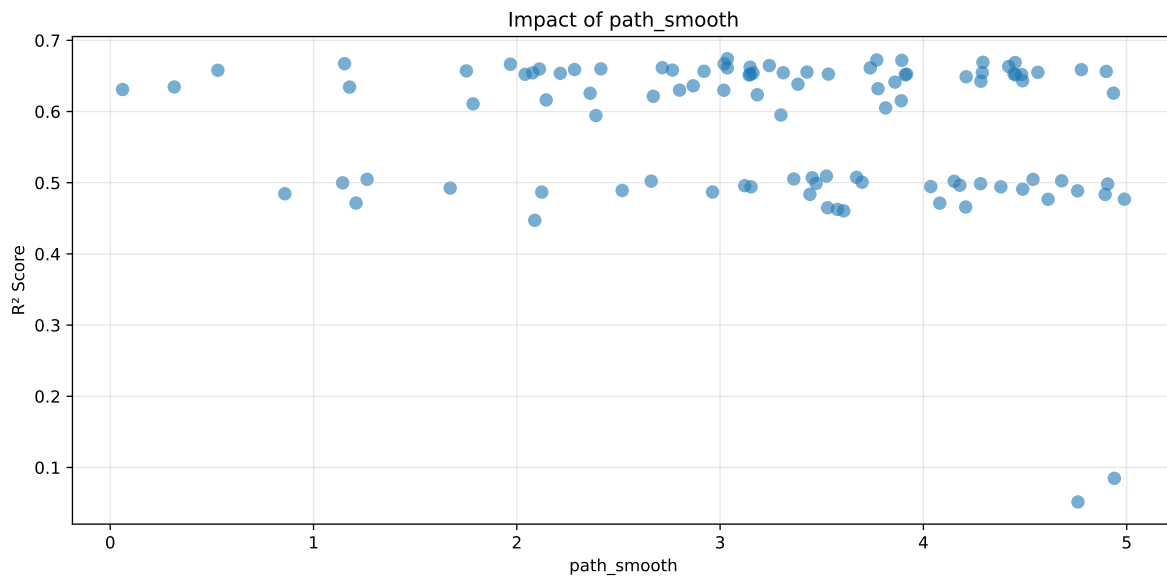
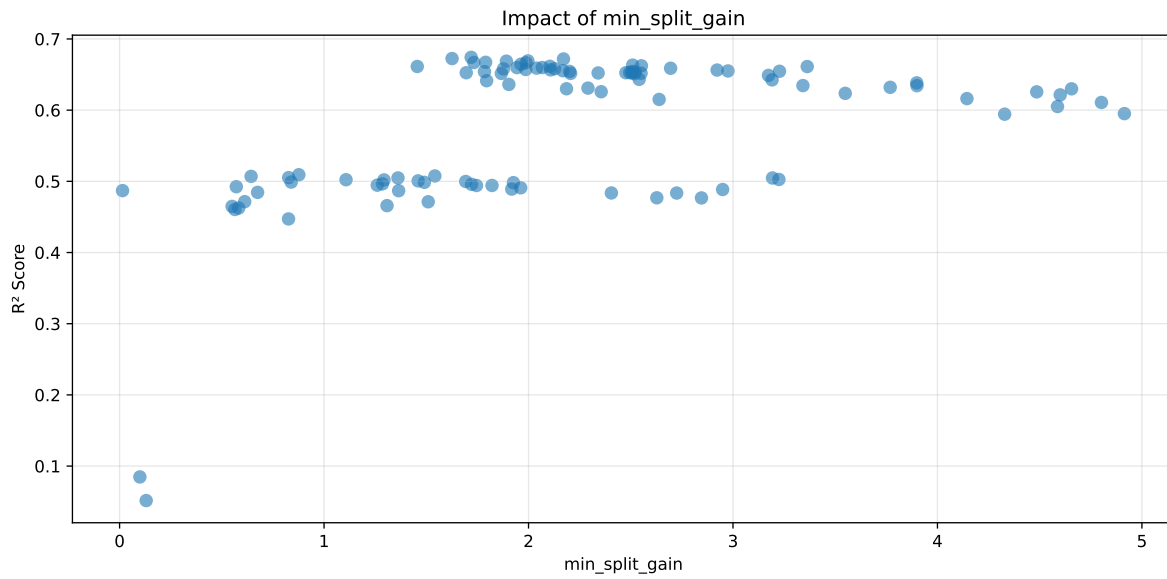
Distribution y_{train} vs y_{test} : y_{train} : mean=2.354, std=0.811, min=0.959, max=13.845
 y_{test} : mean=2.358, std=0.777, min=1.068, max=7.822

===== TOP Importance FEATURES =====

feature	importance
r_CO2	348
r_NH3_st_8h_max	221
r_umidity	186
r_temperature_st_8h_min	111
r_temperature_st_2h_diff_last	99
r_temperature_st_2h_std	96
r_NH3	91
r_umidity_st_2h_slope	84
r_temperature_st_8h_slope	75
r_NH3_st_2h_std	67
r_NH3_st_8h_slope	66
r_temperature	55
r_NH3_st_8h_std	50
r_NH3_st_2h_diff_last	48
r_THI	44
r_temperature_st_8h_std	43
r_NH3_st_2h_slope	41
r_temperature_st_8h_diff_last	31
r_umidity_st_2h_diff_last	30
r_NH3_st_2h_max	30
r_umidity_st_2h_std	24
r_temperature_st_2h_mean	17
r_NH3_st_8h_diff_last	11
r_umidity_st_8h_diff_last	3

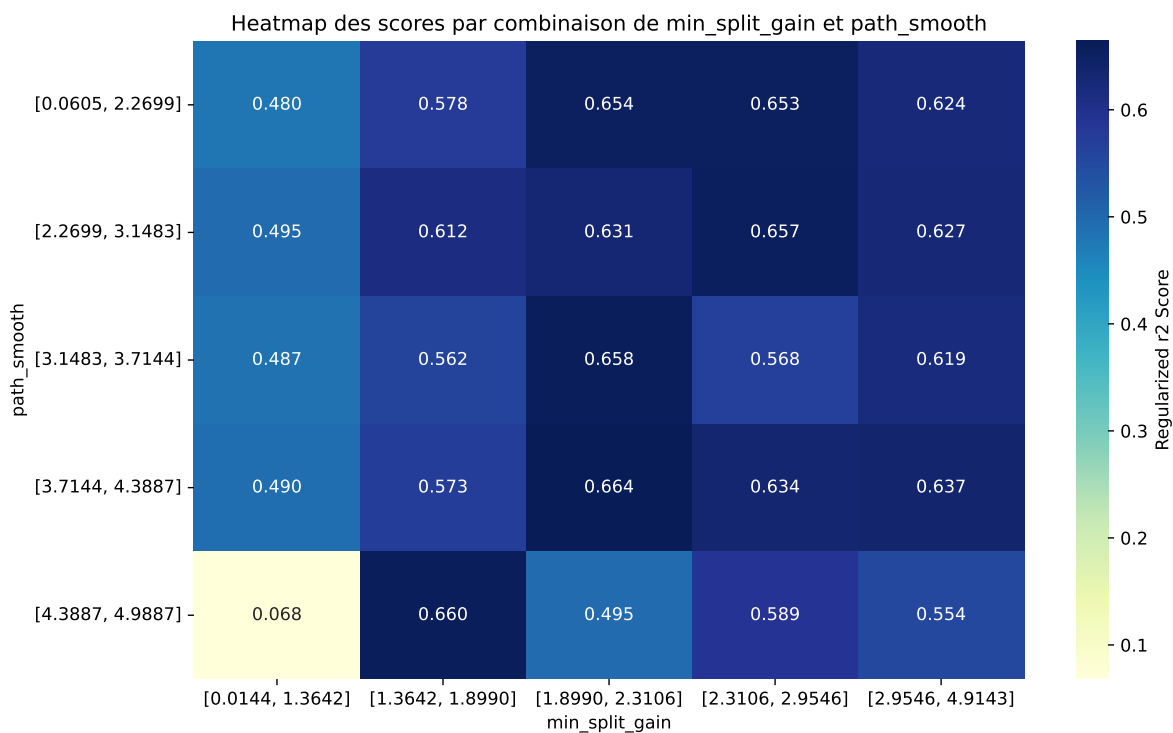


Paramètres avec >5% d'importance : ['min_split_gain', 'path_smooth']



C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of `observed=False` is deprecated and will change to `observed=True` in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7074	0.6742	0.7055	0.0175	0.6742	0.7278	0.0536	1.0795	1.0492
2	0.7053	0.6725	0.7039	0.0157	0.6725	0.7297	0.0572	1.085	1.0488
3	0.7054	0.6719	0.7027	0.0168	0.6719	0.7252	0.0533	1.0793	1.0499
4	0.6975	0.6692	0.7	0.0175	0.6692	0.7232	0.0539	1.0806	1.0423
5	0.7018	0.6688	0.6999	0.0158	0.6688	0.7251	0.0562	1.0841	1.0493
6	0.7004	0.6672	0.6998	0.0173	0.6672	0.7249	0.0577	1.0865	1.0497
7	0.6954	0.6669	0.6988	0.0166	0.6669	0.7225	0.0556	1.0834	1.0427
8	0.6995	0.6665	0.6968	0.0167	0.6665	0.722	0.0555	1.0832	1.0495
9	0.6969	0.6645	0.6951	0.0182	0.6645	0.7181	0.0536	1.0807	1.0488
10	0.6953	0.6633	0.693	0.016	0.6633	0.7152	0.0519	1.0783	1.0484
---	---	---	---	---	---	---	---	---	---
100	0.9213	0.7764	0.7933	0.0182	0.0515	0.8089	0.0326	1.042	1.1867

Hyperparameters:

Rank 1: {'reg__n_estimators': 600, 'reg__max_depth': 8, 'reg__learning_rate': 0.03525230445124458, 'reg__num_leaves': 29, 'reg__subsample': 0.874873559434278, 'reg__colsample_bytree': 0.8250138009846032, 'reg__min_child_samples': 46, 'reg__reg_alpha': 3.8581791559922074, 'reg__reg_lambda': 34.24989394310262, 'reg__min_split_gain': 1.7195184699741028, 'reg__path_smooth': 3.0356152305659014, 'reg__min_child_weight':

0.0010472211291372411, 'reg__feature_fraction': 0.7734183012292501, 'reg__bagging_fraction': 0.8372974087820476, 'reg__bagging_freq': 5, 'reg__max_bin': 159, 'reg__min_data_per_group': 105}

Rank 2: {'reg__n_estimators': 400, 'reg__max_depth': 8, 'reg__learning_rate': 0.023232006846710975, 'reg__num_leaves': 38, 'reg__subsample': 0.6807808723200083, 'reg__colsample_bytree': 0.7411775863034213, 'reg__min_child_samples': 41, 'reg__reg_alpha': 4.053098899459003, 'reg__reg_lambda': 37.58236109465524, 'reg__min_split_gain': 1.6261134279786047, 'reg__path_smooth': 3.770532613414418, 'reg__min_child_weight': 9.551615657443724, 'reg__feature_fraction': 0.5592423524722878, 'reg__bagging_fraction': 0.802094058555559, 'reg__bagging_freq': 2, 'reg__max_bin': 230, 'reg__min_data_per_group': 133}

Rank 3: {'reg__n_estimators': 600, 'reg__max_depth': 7, 'reg__learning_rate': 0.02867343286690967, 'reg__num_leaves': 35, 'reg__subsample': 0.6269940027841847, 'reg__colsample_bytree': 0.7872976638460523, 'reg__min_child_samples': 36, 'reg__reg_alpha': 1.3218743369775834, 'reg__reg_lambda': 32.937532088907076, 'reg__min_split_gain': 2.1709790972386656, 'reg__path_smooth': 3.893788930682981, 'reg__min_child_weight': 0.08447351693383155, 'reg__feature_fraction': 0.6195678019375457, 'reg__bagging_fraction': 0.8684699767949915, 'reg__bagging_freq': 5, 'reg__max_bin': 82, 'reg__min_data_per_group': 121}

Rank 4: {'reg__n_estimators': 800, 'reg__max_depth': 6, 'reg__learning_rate': 0.05314334790173279, 'reg__num_leaves': 54, 'reg__subsample': 0.8934615070786254, 'reg__colsample_bytree': 0.6116574246532208, 'reg__min_child_samples': 25, 'reg__reg_alpha': 3.078091074561334, 'reg__reg_lambda': 49.21770740817692, 'reg__min_split_gain': 1.9969099630063338, 'reg__path_smooth': 4.2927361974370655, 'reg__min_child_weight': 6.506334995498096, 'reg__feature_fraction': 0.566000508844331, 'reg__bagging_fraction': 0.6168590699953299, 'reg__bagging_freq': 1, 'reg__max_bin': 195, 'reg__min_data_per_group': 51}

Rank 5: {'reg__n_estimators': 500, 'reg__max_depth': 5, 'reg__learning_rate': 0.03979046479392884, 'reg__num_leaves': 31, 'reg__subsample': 0.8617477300128145, 'reg__colsample_bytree': 0.6636938013021995, 'reg__min_child_samples': 33, 'reg__reg_alpha': 2.314911761809546, 'reg__reg_lambda': 36.99825404377027, 'reg__min_split_gain': 1.8918402609351221, 'reg__path_smooth': 4.451487640047633, 'reg__min_child_weight': 0.07053203535491803, 'reg__feature_fraction': 0.5907005194572759, 'reg__bagging_fraction': 0.817983311994098, 'reg__bagging_freq': 3, 'reg__max_bin': 115, 'reg__min_data_per_group': 67}

Rank 6: {'reg__n_estimators': 600, 'reg__max_depth': 8, 'reg__learning_rate': 0.021199575563034506, 'reg__num_leaves': 43, 'reg__subsample': 0.6896690346112506, 'reg__colsample_bytree': 0.7247114226442588, 'reg__min_child_samples': 44, 'reg__reg_alpha': 4.1982317350885445, 'reg__reg_lambda': 45.05926031202917, 'reg__min_split_gain': 1.7902676365571442, 'reg__path_smooth': 1.1526217763225926, 'reg__min_child_weight':

0.007810836815870446, 'reg__feature_fraction': 0.6869530944253763, 'reg__bagging_fraction': 0.862228795787812, 'reg__bagging_freq': 5, 'reg__max_bin': 203, 'reg__min_data_per_group': 137}

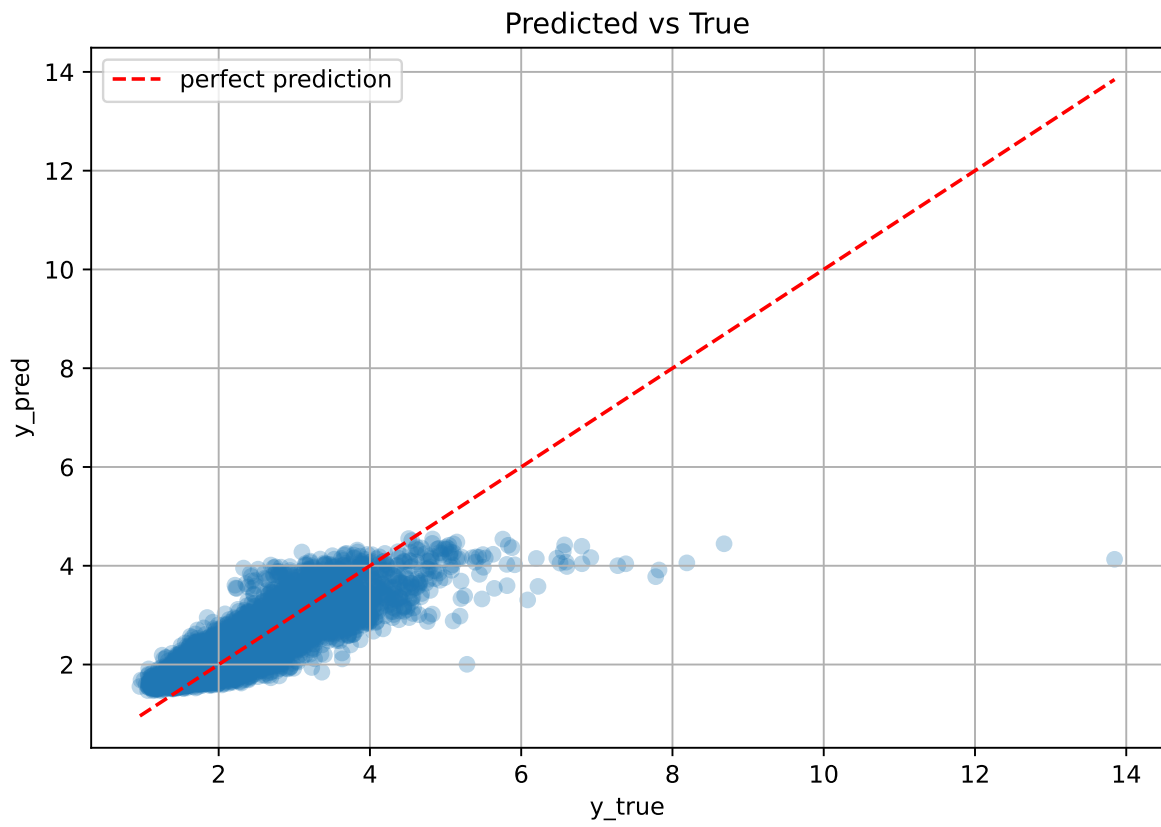
Rank 7: {'reg__n_estimators': 600, 'reg__max_depth': 8, 'reg__learning_rate': 0.03341592616490544, 'reg__num_leaves': 62, 'reg__subsample': 0.8672830760768173, 'reg__colsample_bytree': 0.8270886488680151, 'reg__min_child_samples': 46, 'reg__reg_alpha': 5.172216910811437, 'reg__reg_lambda': 31.85183880338135, 'reg__min_split_gain': 1.7328607738285515, 'reg__path_smooth': 3.0199417938116437, 'reg__min_child_weight': 0.08014817255584815, 'reg__feature_fraction': 0.5012654541029933, 'reg__bagging_fraction': 0.882247500373903, 'reg__bagging_freq': 5, 'reg__max_bin': 93, 'reg__min_data_per_group': 125}

Rank 8: {'reg__n_estimators': 300, 'reg__max_depth': 5, 'reg__learning_rate': 0.04330981916727886, 'reg__num_leaves': 59, 'reg__subsample': 0.8267509211231444, 'reg__colsample_bytree': 0.8603556134666542, 'reg__min_child_samples': 46, 'reg__reg_alpha': 1.628268520174944, 'reg__reg_lambda': 26.29497219057405, 'reg__min_split_gain': 1.9878344559833891, 'reg__path_smooth': 1.9685720688478336, 'reg__min_child_weight': 6.330173184930593, 'reg__feature_fraction': 0.8013973165593594, 'reg__bagging_fraction': 0.7832895071316183, 'reg__bagging_freq': 6, 'reg__max_bin': 95, 'reg__min_data_per_group': 113}

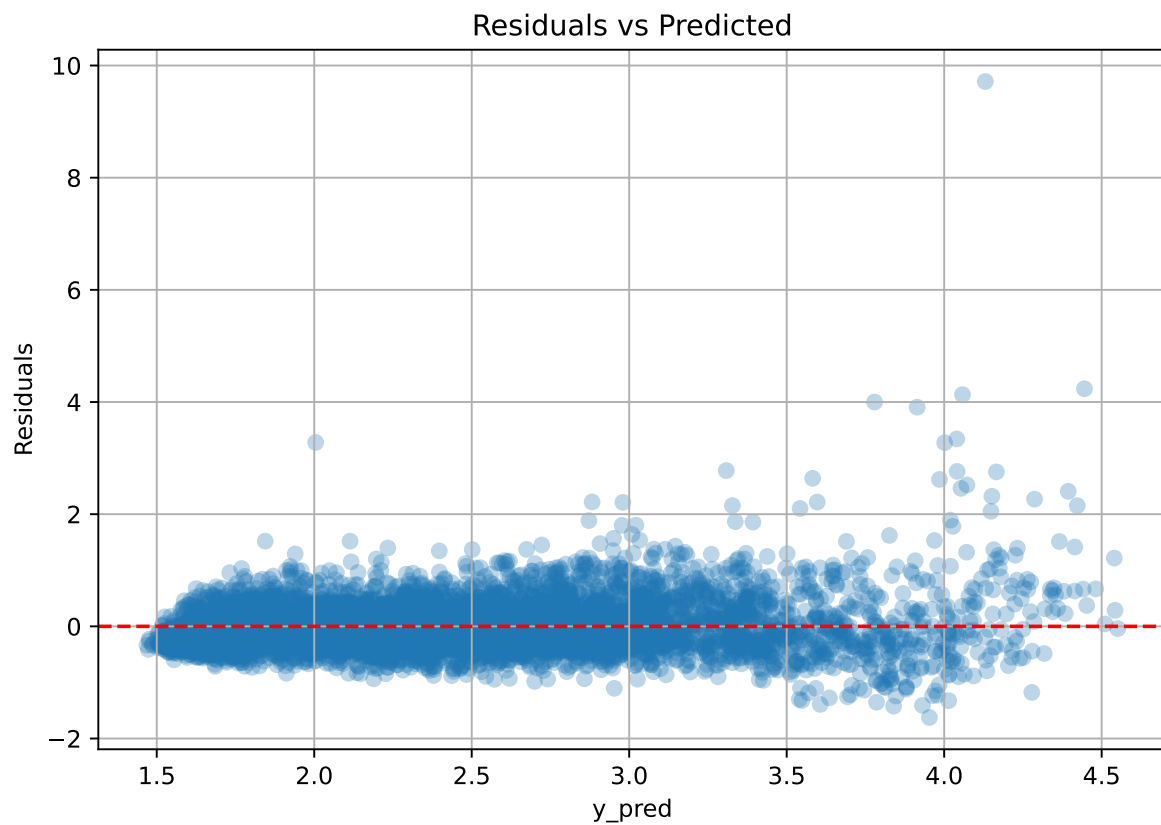
Rank 9: {'reg__n_estimators': 200, 'reg__max_depth': 8, 'reg__learning_rate': 0.016660387693083218, 'reg__num_leaves': 22, 'reg__subsample': 0.8810218902962609, 'reg__colsample_bytree': 0.8889314798185185, 'reg__min_child_samples': 47, 'reg__reg_alpha': 1.730303306921062, 'reg__reg_lambda': 26.53406607955442, 'reg__min_split_gain': 1.9628409122675674, 'reg__path_smooth': 3.2421192383884523, 'reg__min_child_weight': 0.0014441817861399124, 'reg__feature_fraction': 0.7636124425953958, 'reg__bagging_fraction': 0.8476076091015395, 'reg__bagging_freq': 6, 'reg__max_bin': 149, 'reg__min_data_per_group': 95}

Rank 10: {'reg__n_estimators': 300, 'reg__max_depth': 8, 'reg__learning_rate': 0.017367162350111195, 'reg__num_leaves': 30, 'reg__subsample': 0.6041666210916475, 'reg__colsample_bytree': 0.7539254971290968, 'reg__min_child_samples': 32, 'reg__reg_alpha': 1.5353256388869052, 'reg__reg_lambda': 28.176042667266017, 'reg__min_split_gain': 2.5091869828960167, 'reg__path_smooth': 4.419749657258984, 'reg__min_child_weight': 0.0036097936289552573, 'reg__feature_fraction': 0.5883843789477009, 'reg__bagging_fraction': 0.8369541668384891, 'reg__bagging_freq': 5, 'reg__max_bin': 120, 'reg__min_data_per_group': 102}

3.5.2 Scatter



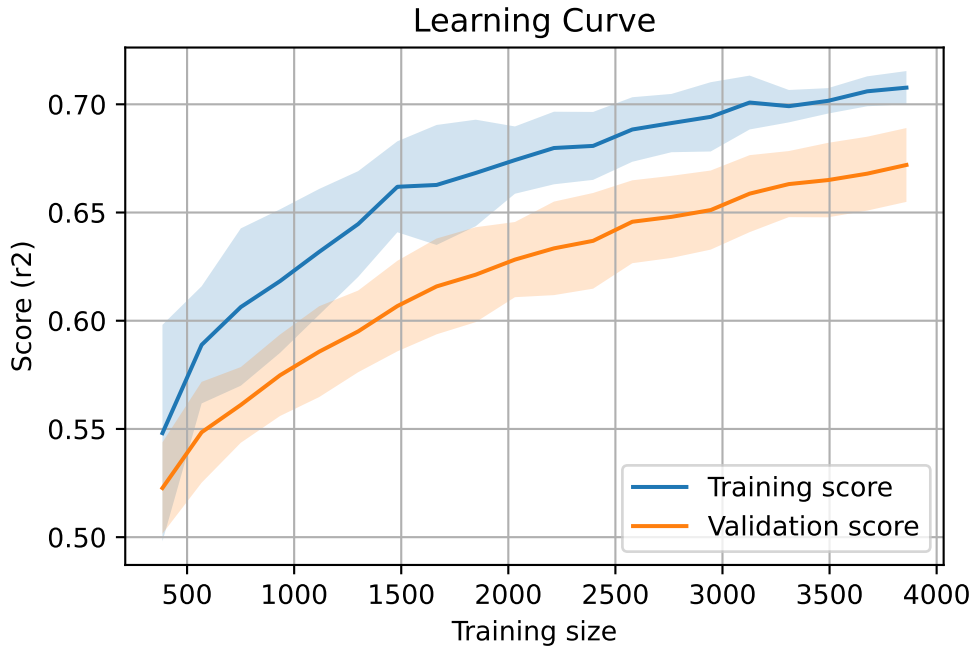
3.5.3 Scatter residuals



3.5.4 Learning curve

Len(group) : 7876, **Len(training)** : 5514,

Len(training_real) : 3860, **Len(test_of_training)** : 1654, **Len(test)** : 2362

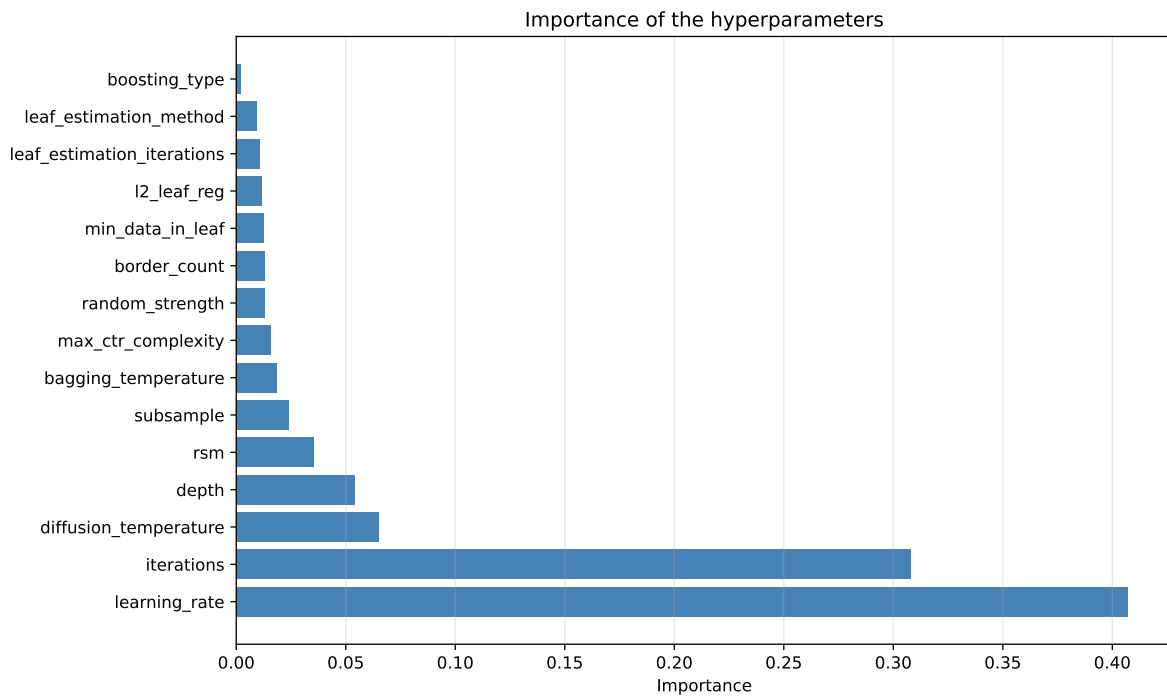
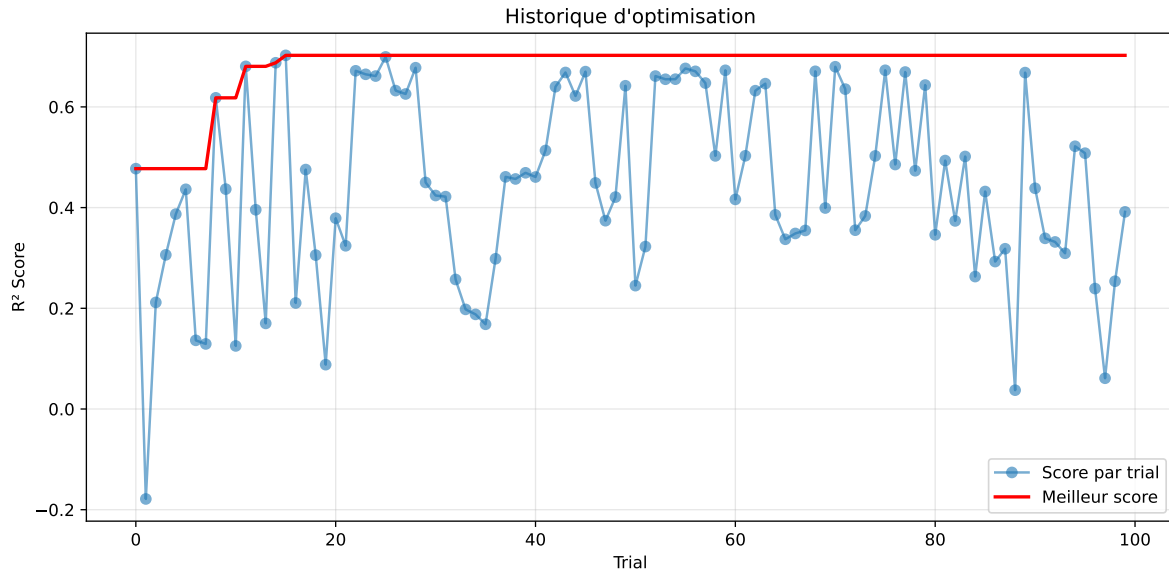


3.6 Model : CatBoost

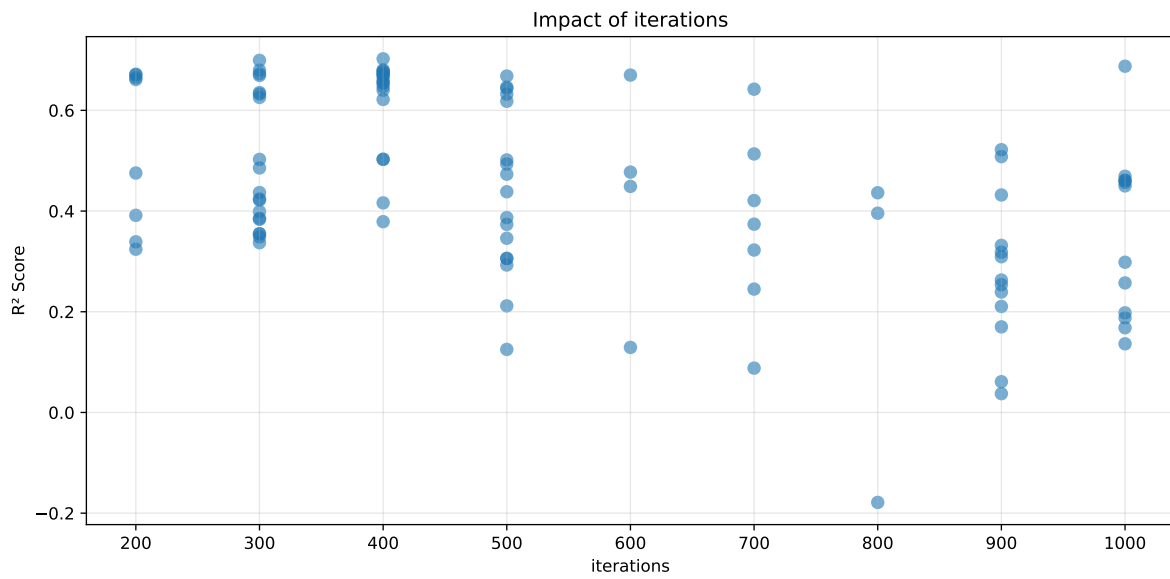
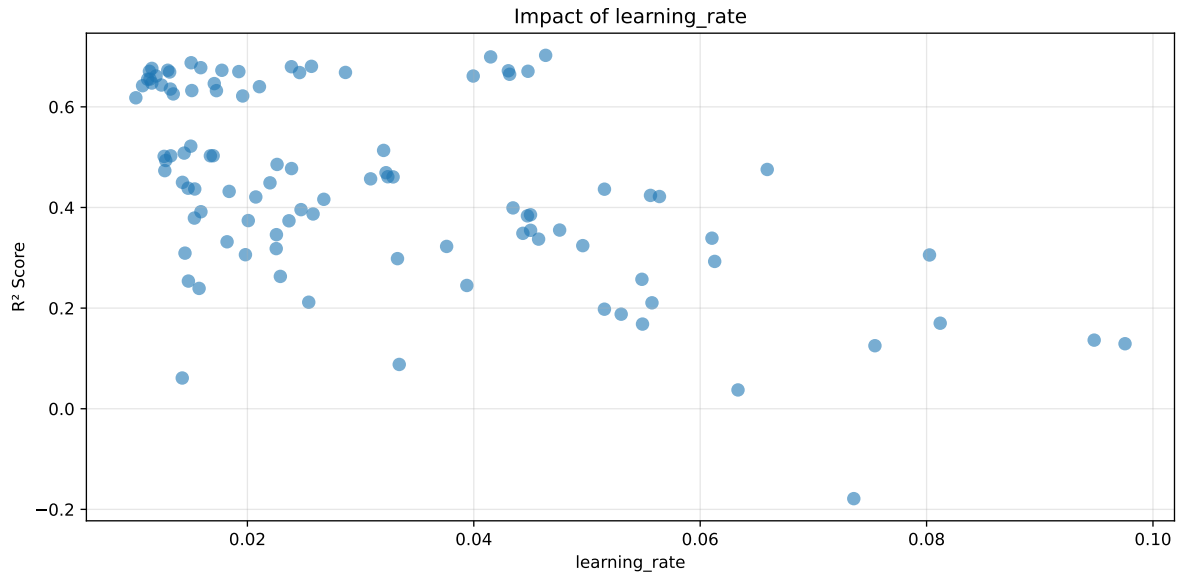
3.6.1 Gridsearch

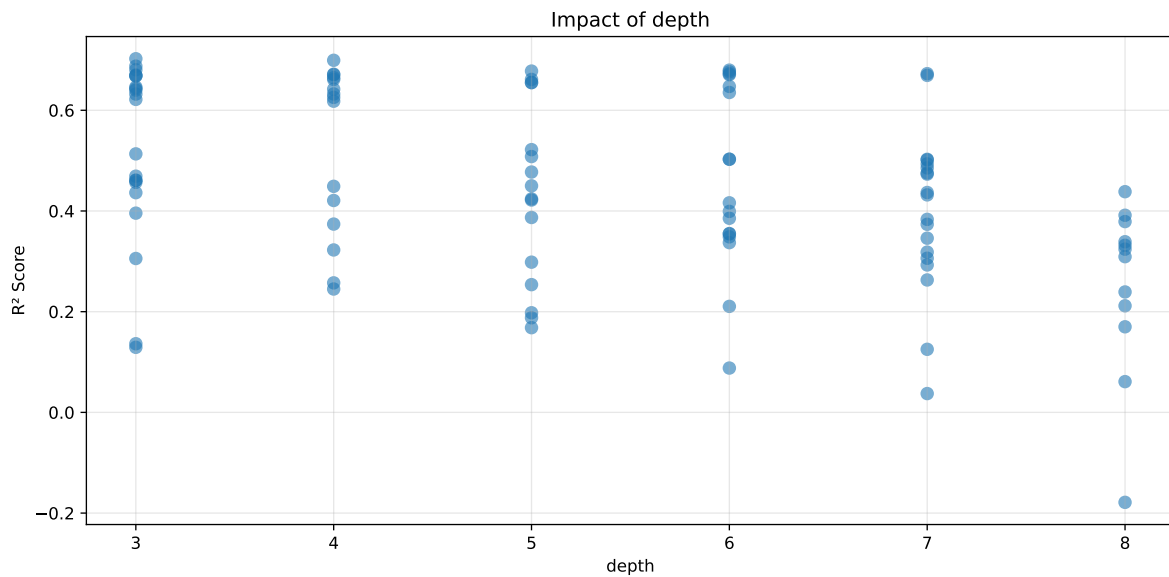
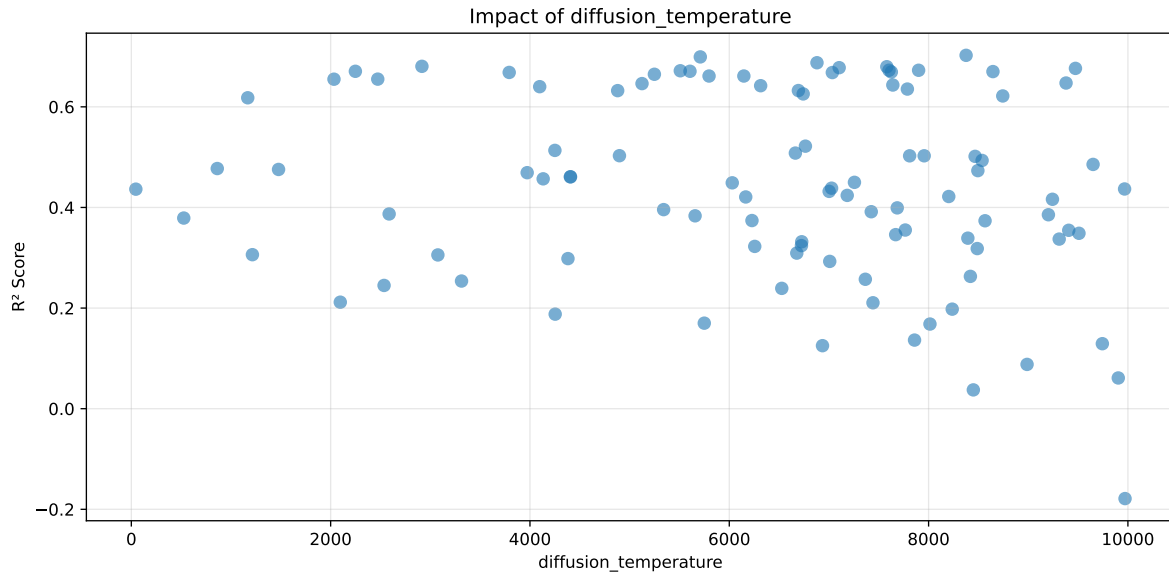
Distribution y_train vs y_test: y_train: mean=2.354, std=0.811, min=0.959, max=13.845
y_test: mean=2.358, std=0.777, min=1.068, max=7.822

```
===== TOP Importance FEATURES =====
feature importance r_umidity 17.998374 r_umidity_st_2h_slope 9.614321 r_temper-
ature_st_2h_std 9.387163 r_CO2 8.204981 r_temperature_st_8h_slope 8.082962
r_NH3_st_8h_max 6.201936 r_temperature_st_8h_diff_last 5.184821 r_NH3_st_2h_std
4.312118 r_temperature_st_8h_min 3.933761 r_temperature_st_2h_diff_last 3.883547
r_NH3 3.733110 r_temperature 3.459063 r_THI 2.564857 r_NH3_st_8h_std 2.440928
r_NH3_st_2h_max 2.134504 r_temperature_st_2h_mean 1.786637 r_NH3_st_8h_slope
1.410633 r_NH3_st_2h_slope 1.395759 r_temperature_st_8h_std 1.066663 r_umid-
ity_st_2h_std 0.817357 r_umidity_st_8h_diff_last 0.710127 r_NH3_st_2h_diff_last
0.601552 r_NH3_st_8h_diff_last 0.541488 r_umidity_st_2h_diff_last 0.533339
```



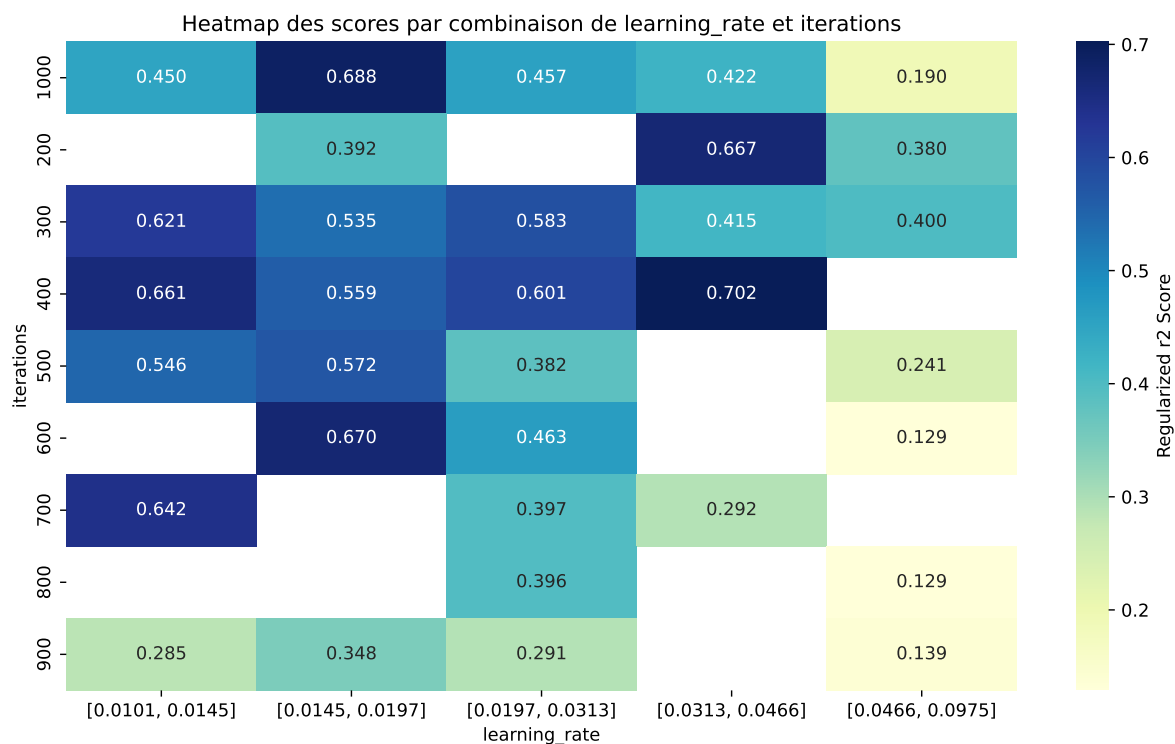
Paramètres avec >5% d'importance : ['learning_rate', 'iterations', 'diffusion_temperature', 'depth']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7364	0.7024	0.7302	0.0188	0.7024	0.7413	0.0389	1.0554	1.0485
2	0.7324	0.6993	0.7274	0.0196	0.6993	0.7355	0.0363	1.0519	1.0474
3	0.714	0.6876	0.7188	0.0179	0.6876	0.7255	0.0379	1.0551	1.0383
4	0.714	0.6805	0.7097	0.0167	0.6805	0.7162	0.0357	1.0525	1.0492
5	0.7114	0.6798	0.7066	0.017	0.6798	0.72	0.0403	1.0593	1.0465
6	0.705	0.6778	0.7059	0.016	0.6778	0.7148	0.037	1.0546	1.04
7	0.706	0.6764	0.7036	0.017	0.6764	0.7107	0.0342	1.0506	1.0437
8	0.7048	0.6727	0.7005	0.0156	0.6727	0.7133	0.0405	1.0603	1.0477
9	0.7052	0.6727	0.7	0.0159	0.6727	0.711	0.0383	1.057	1.0483
10	0.6937	0.6717	0.7001	0.0153	0.6717	0.7018	0.0301	1.0449	1.0327
---	---	---	---	---	---	---	---	---	---
100	0.9951	0.7994	0.8096	0.0246	-0.1787	0.8296	0.0301	1.0377	1.2447

Hyperparameters:

Rank 1: {'reg__iterations': 400, 'reg__depth': 3, 'reg__learning_rate': 0.04634690900600413, 'reg__l2_leaf_reg': 17.143938698017273, 'reg__bagging_temperature': 4.665157499978544, 'reg__random_strength': 2.8265414036233114, 'reg__subsample': 0.6032543962928424, 'reg__min_data_in_leaf': 40, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.5565497317752494, 'reg__border_count': 110, 'reg__leaf_estimation_method': 'Newton',

'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 8376.347451152087}

Rank 2: {'reg__iterations': 300, 'reg__depth': 4, 'reg__learning_rate': 0.04148771218882057, 'reg__l2_leaf_reg': 18.430199546589595, 'reg__bagging_temperature': 2.095282048458126, 'reg__random_strength': 2.2229757121572327, 'reg__subsample': 0.5025115574857266, 'reg__min_data_in_leaf': 24, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.504141065522266, 'reg__border_count': 97, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 5709.237819079772}

Rank 3: {'reg__iterations': 1000, 'reg__depth': 3, 'reg__learning_rate': 0.01502888622498196, 'reg__l2_leaf_reg': 23.831826460385418, 'reg__bagging_temperature': 0.7814567602047318, 'reg__random_strength': 3.4074276180069774, 'reg__subsample': 0.5457478025233323, 'reg__min_data_in_leaf': 47, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.8720226309475217, 'reg__border_count': 86, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 6880.02833717138}

Rank 4: {'reg__iterations': 400, 'reg__depth': 3, 'reg__learning_rate': 0.025657323283017338, 'reg__l2_leaf_reg': 8.773470181653419, 'reg__bagging_temperature': 1.419642431512872, 'reg__random_strength': 0.5486844442057448, 'reg__subsample': 0.8219491813335525, 'reg__min_data_in_leaf': 19, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.6206970496728057, 'reg__border_count': 75, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 2915.9211734092705}

Rank 5: {'reg__iterations': 300, 'reg__depth': 6, 'reg__learning_rate': 0.023887606140115192, 'reg__l2_leaf_reg': 17.334037758661495, 'reg__bagging_temperature': 1.4493242198552072, 'reg__random_strength': 2.3544093993877704, 'reg__subsample': 0.739252487514458, 'reg__min_data_in_leaf': 32, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.6404992582773451, 'reg__border_count': 79, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 7580.6065222466195}

Rank 6: {'reg__iterations': 400, 'reg__depth': 5, 'reg__learning_rate': 0.015878430354360783, 'reg__l2_leaf_reg': 23.27811646972869, 'reg__bagging_temperature': 2.181658866288729, 'reg__random_strength': 1.2936815469216747, 'reg__subsample': 0.8908847030246729, 'reg__min_data_in_leaf': 25, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.5849137616719458, 'reg__border_count': 127, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 7102.722295222173}

Rank 7: {'reg__iterations': 400, 'reg__depth': 6, 'reg__learning_rate': 0.011558864494164984, 'reg__l2_leaf_reg': 34.68230806577623, 'reg__bagging_temperature': 1.7883492214792625, 'reg__random_strength': 0.939263721004465, 'reg__subsample': 0.8703094580424883,

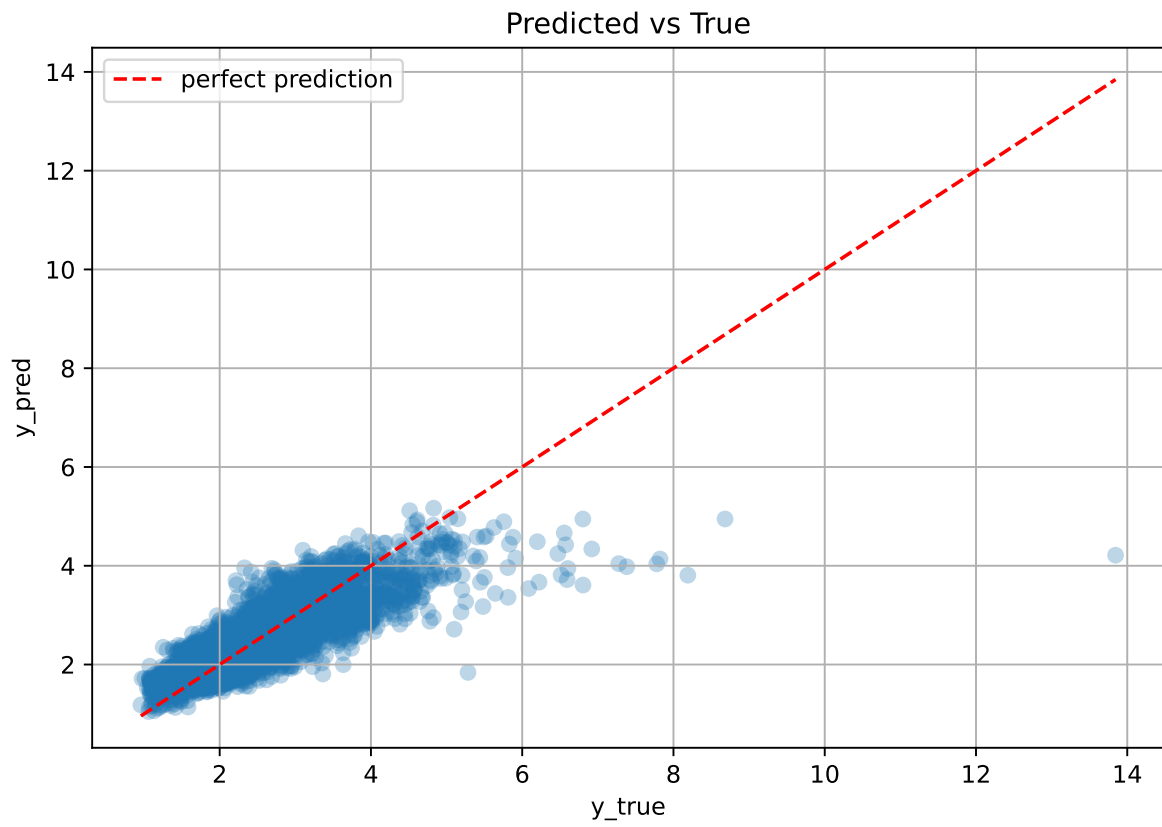
'reg__min_data_in_leaf': 29, 'reg__leaf_estimation_iterations': 9, 'reg__rsm': 0.584926908953573, 'reg__border_count': 104, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 9473.60123129705}

Rank 8: {'reg__iterations': 300, 'reg__depth': 7, 'reg__learning_rate': 0.01296396296646044, 'reg__l2_leaf_reg': 16.540033883505494, 'reg__bagging_temperature': 1.416392372905643, 'reg__random_strength': 0.6053973734459289, 'reg__subsample': 0.8946089714280429, 'reg__min_data_in_leaf': 34, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.6341424247840961, 'reg__border_count': 101, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 7602.809811063485}

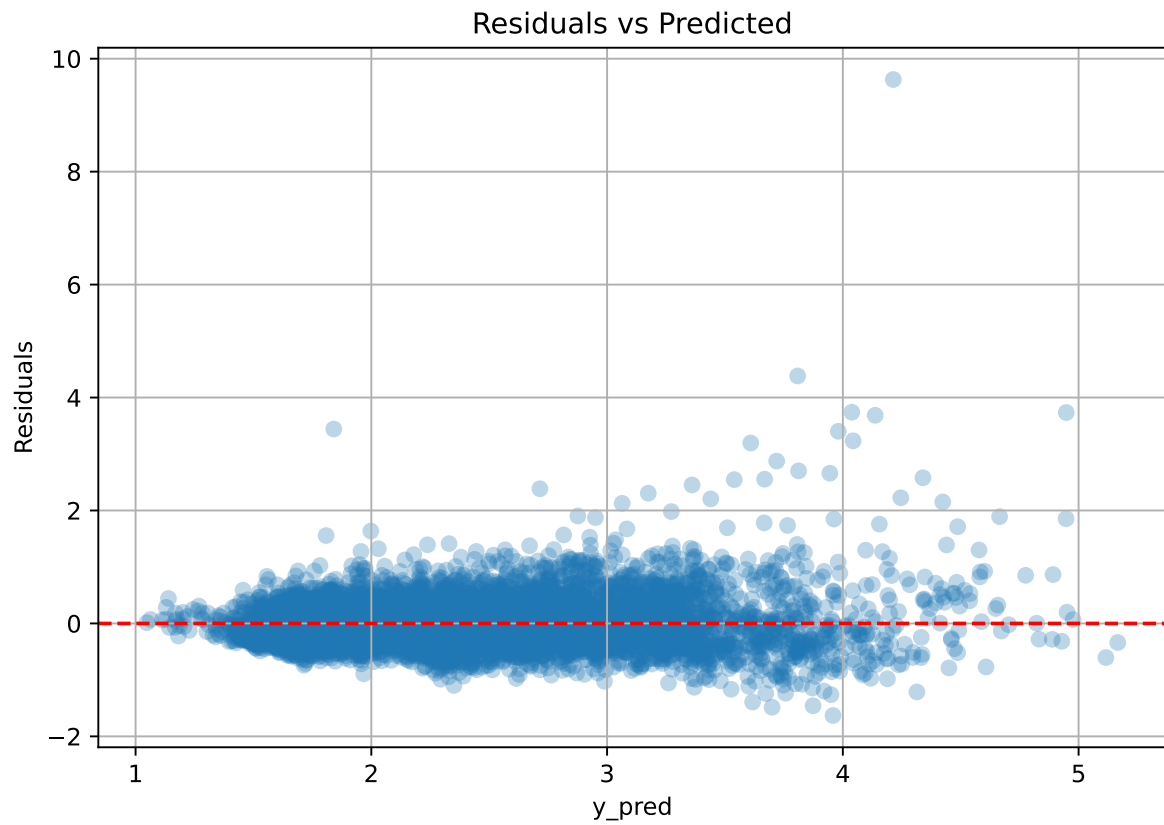
Rank 9: {'reg__iterations': 400, 'reg__depth': 6, 'reg__learning_rate': 0.017744334783116926, 'reg__l2_leaf_reg': 20.728622017088888, 'reg__bagging_temperature': 1.6625051756002875, 'reg__random_strength': 3.2369314560337625, 'reg__subsample': 0.8757620968024402, 'reg__min_data_in_leaf': 29, 'reg__leaf_estimation_iterations': 3, 'reg__rsm': 0.6007392629658534, 'reg__border_count': 92, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 7899.818080830338}

Rank 10: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.04305295547534823, 'reg__l2_leaf_reg': 16.116761765024354, 'reg__bagging_temperature': 4.775435537408709, 'reg__random_strength': 2.0724130417493303, 'reg__subsample': 0.5255252001810213, 'reg__min_data_in_leaf': 19, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.5002180450871931, 'reg__border_count': 96, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 5508.658711040162}

3.6.2 Scatter



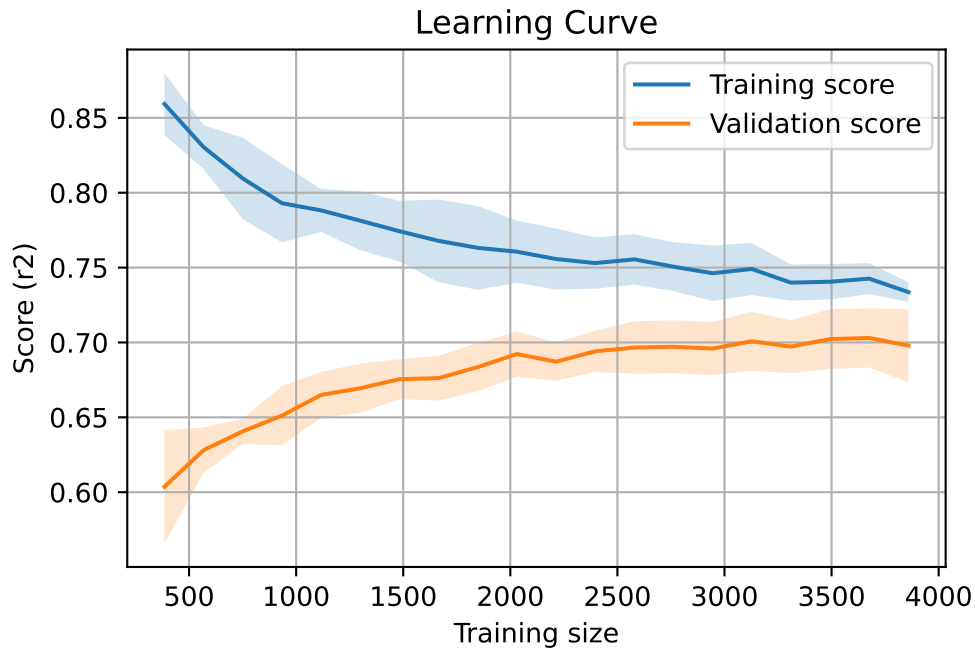
3.6.3 Scatter residuals



3.6.4 Learning curve

`Len(group)` : 7876, `Len(training)` : 5514,

`Len(training_real)` : 3860, `Len(test_of_training)` : 1654, `Len(test)` : 2362



3.7 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.546	0.544	0.015	0.578	0.58	0.036	1.066	1.004	nan
LGBM	0.707	0.674	0.018	0.706	0.728	0.054	1.08	1.049	0.6742
CatBoost	0.736	0.702	0.019	0.73	0.741	0.039	1.055	1.048	0.7024

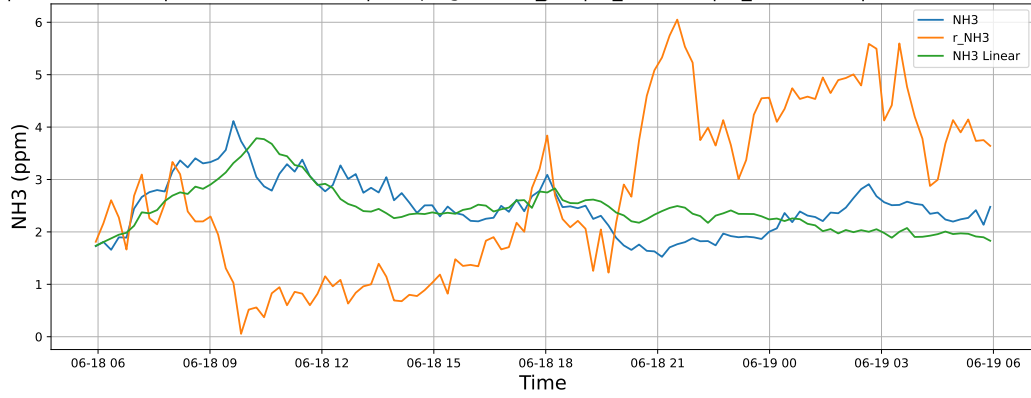
4 Plot

4.1 Standalone plots

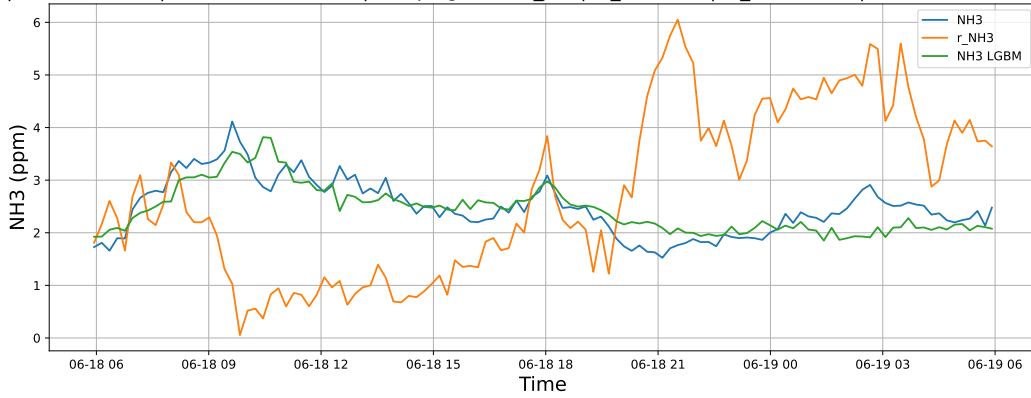
4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

4.1.1.1 Time window : 24

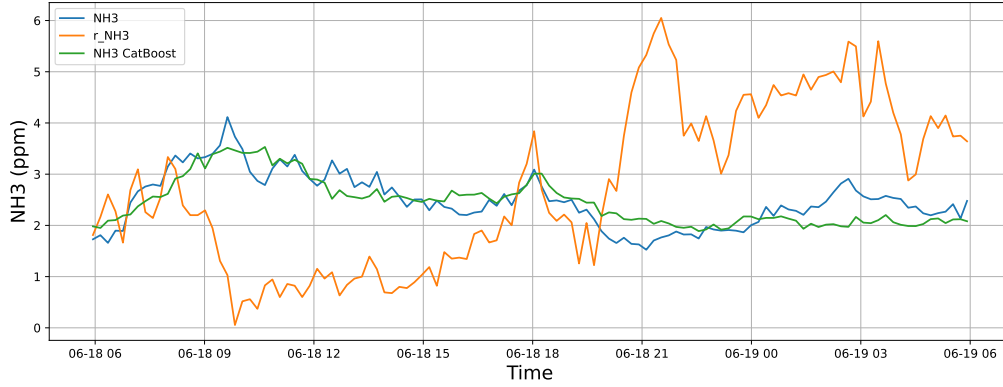
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

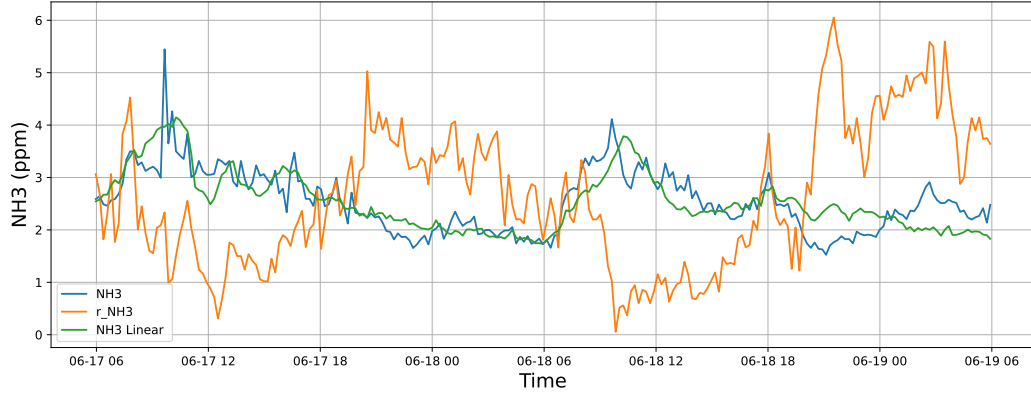


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

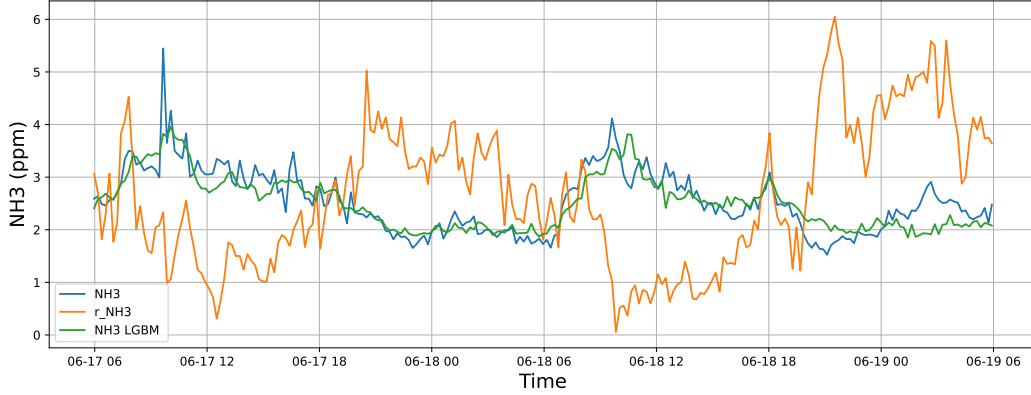


4.1.1.2 Time window : 48

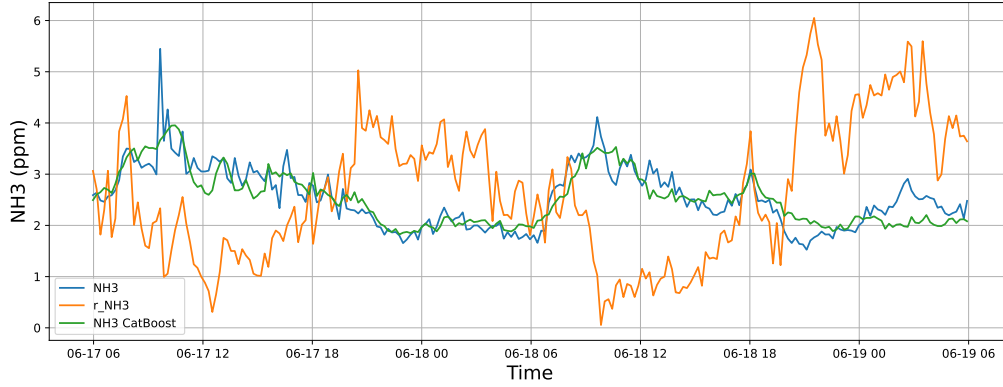
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

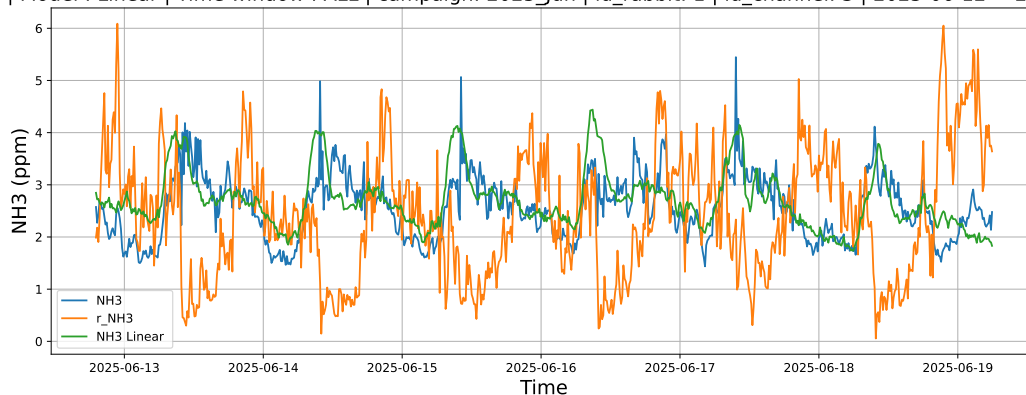


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

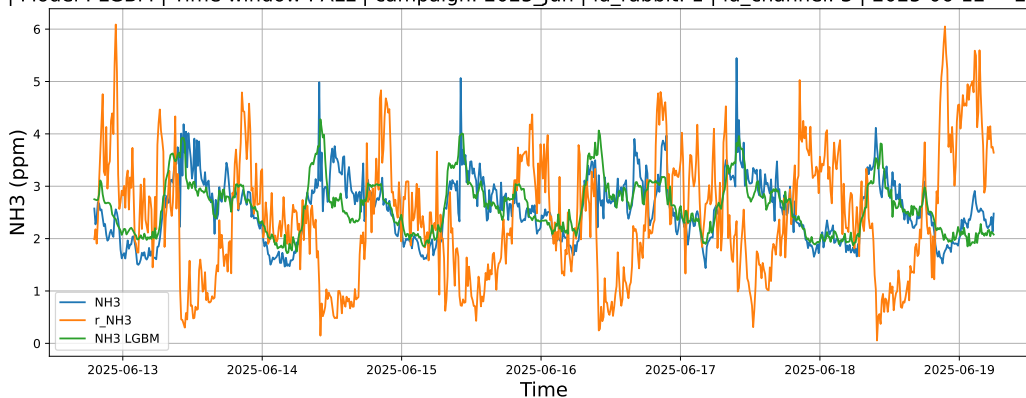


4.1.1.3 Time window : ALL

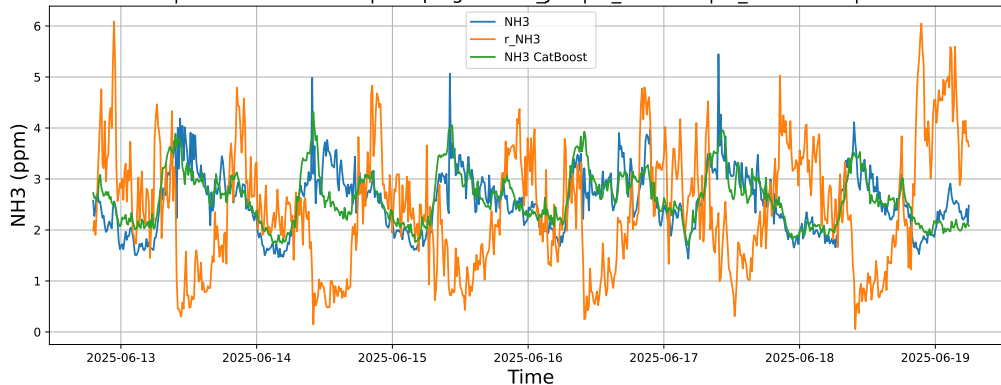
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



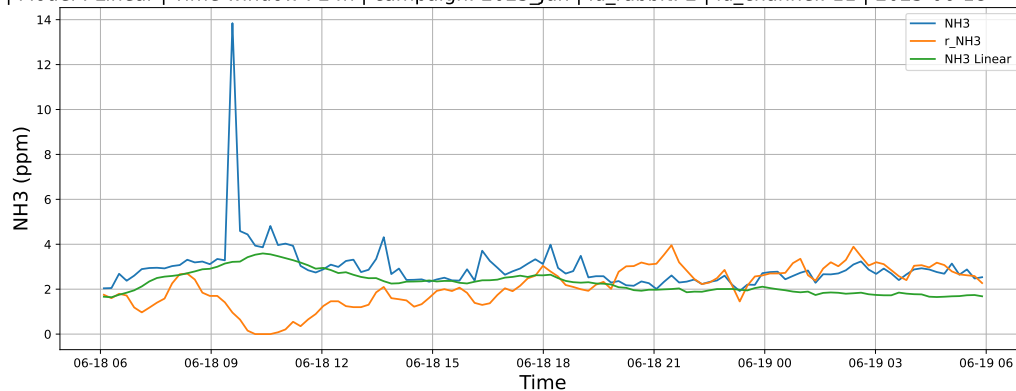
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



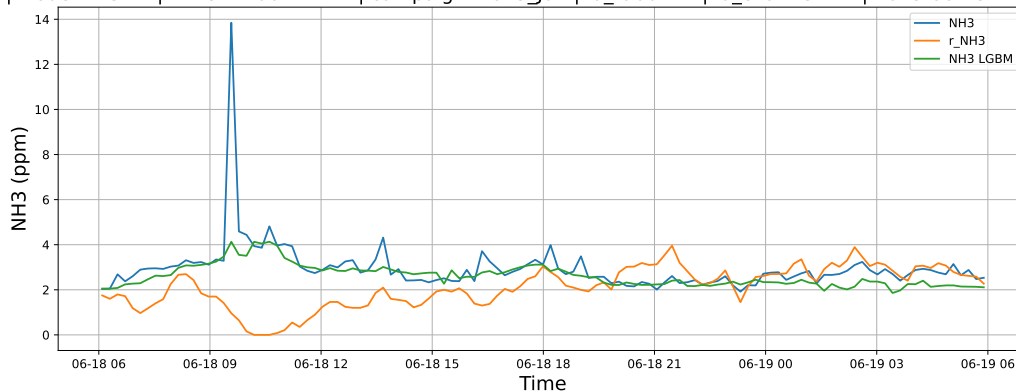
4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

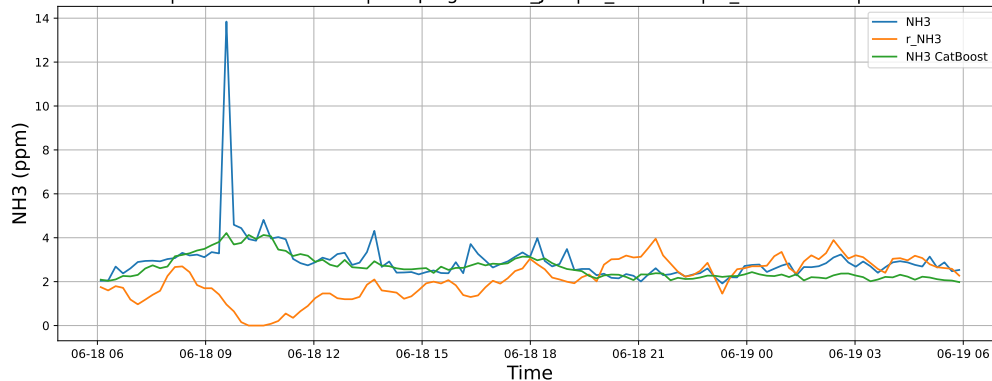
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

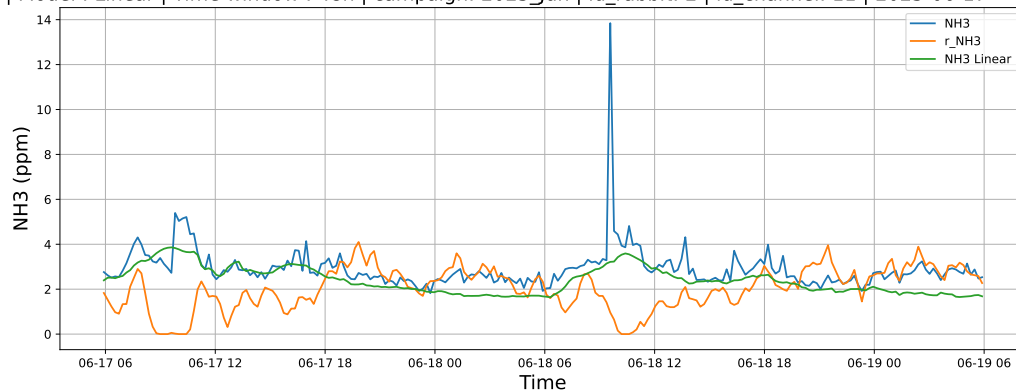


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

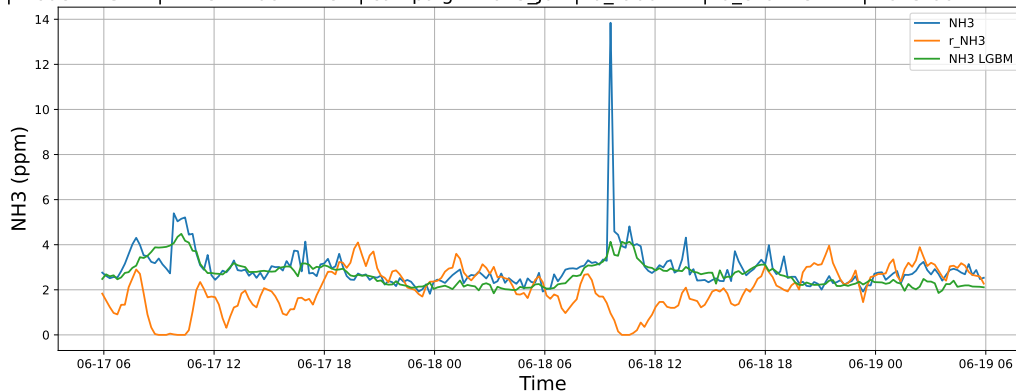


4.1.2.2 Time window : 48

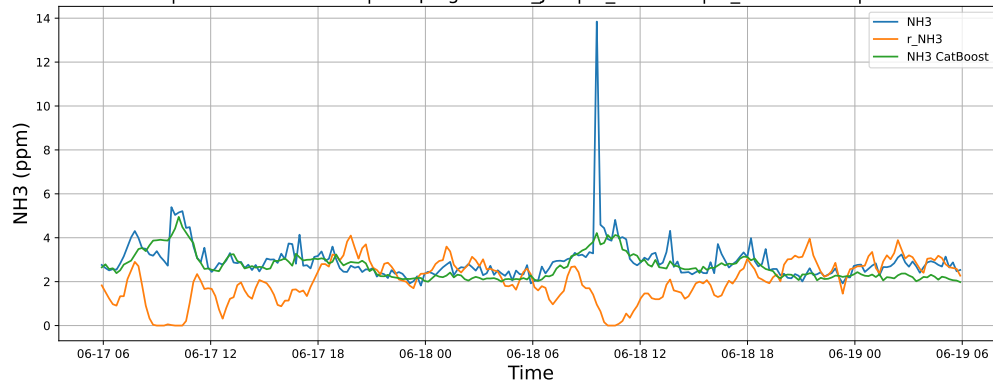
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

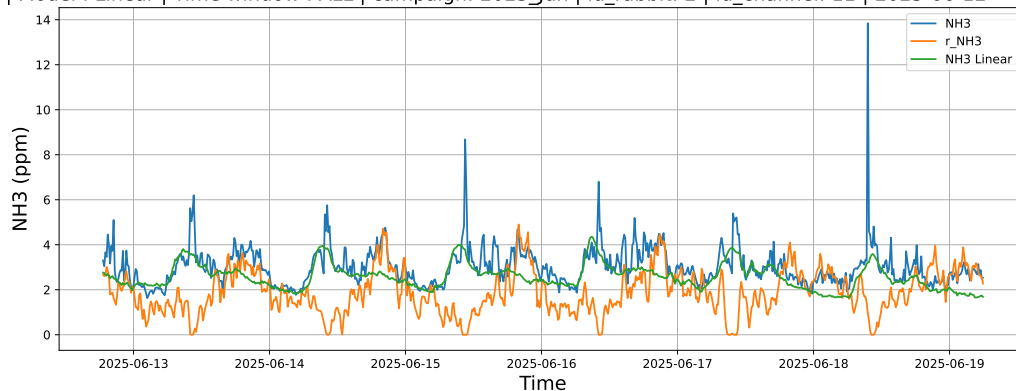


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

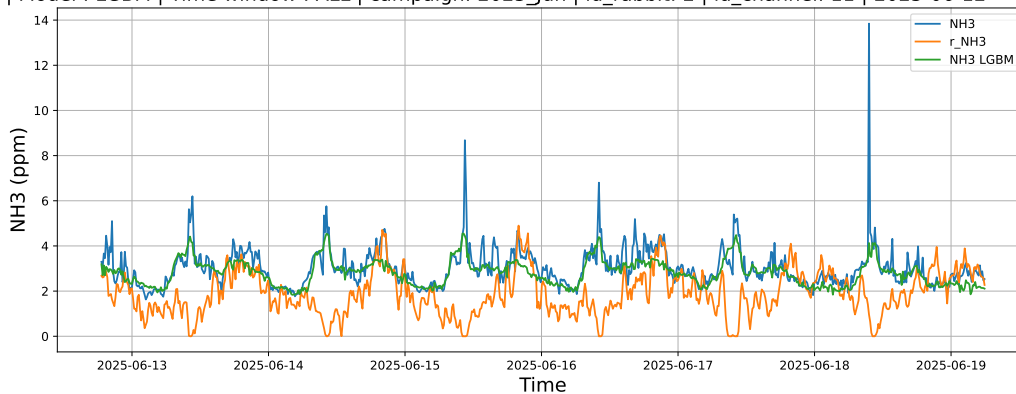


4.1.2.3 Time window : ALL

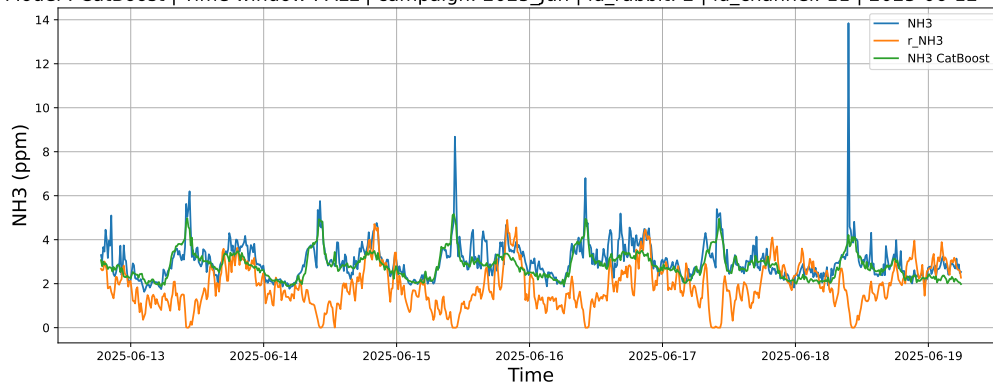
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



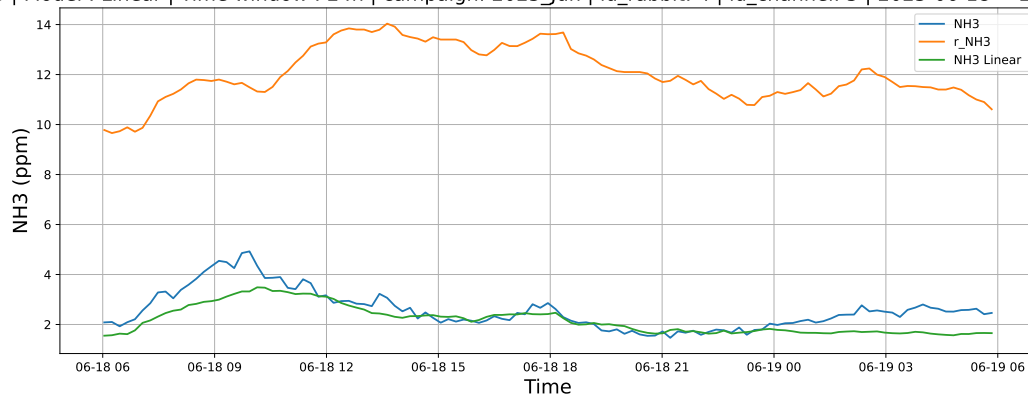
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



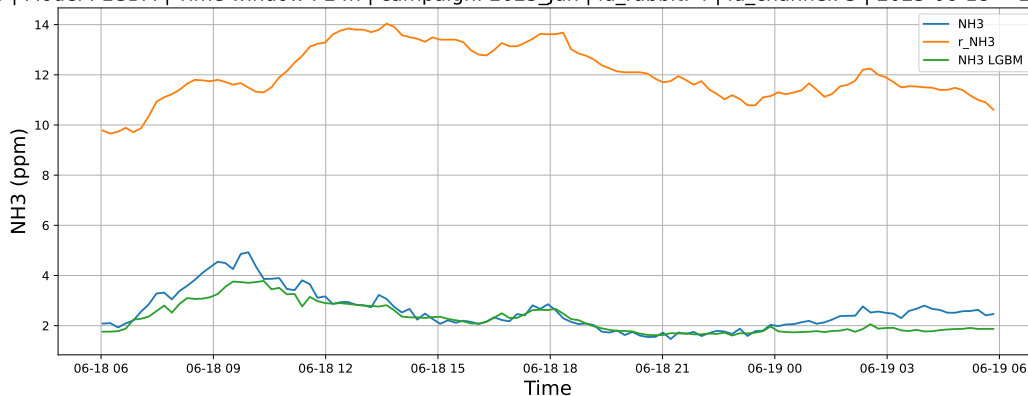
4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.1.3.1 Time window : 24

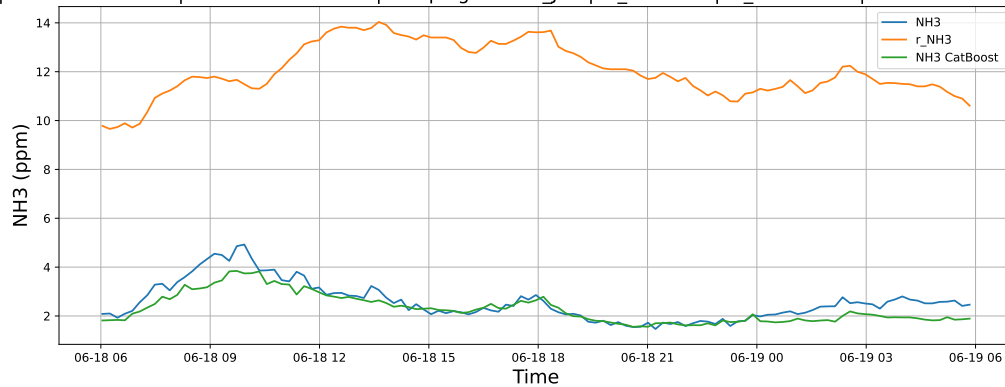
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

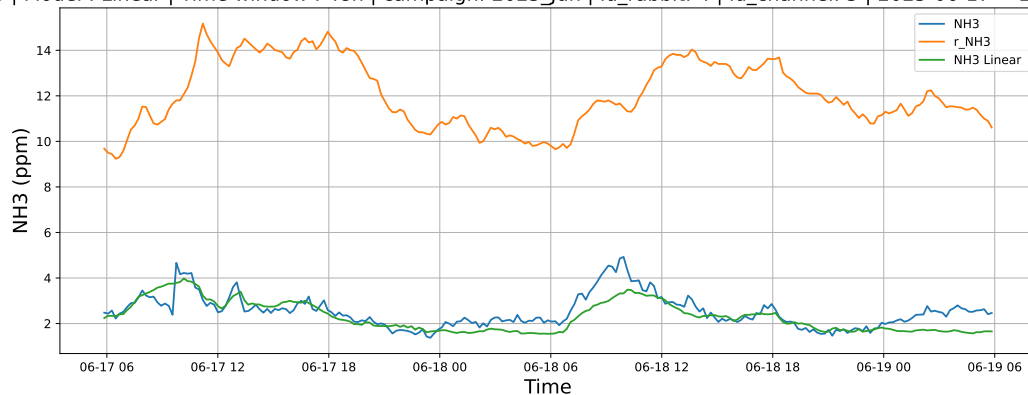


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

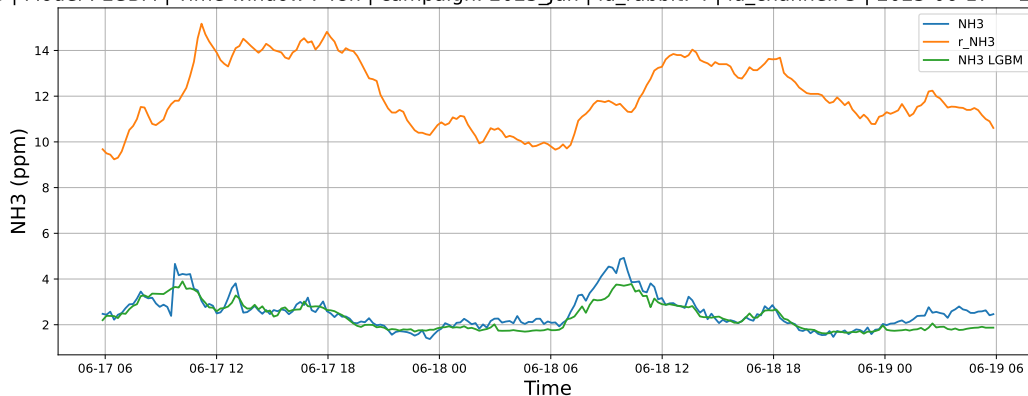


4.1.3.2 Time window : 48

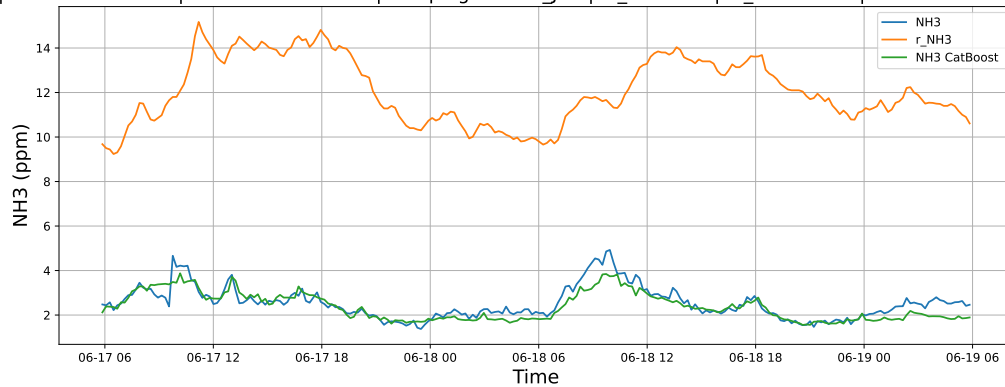
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

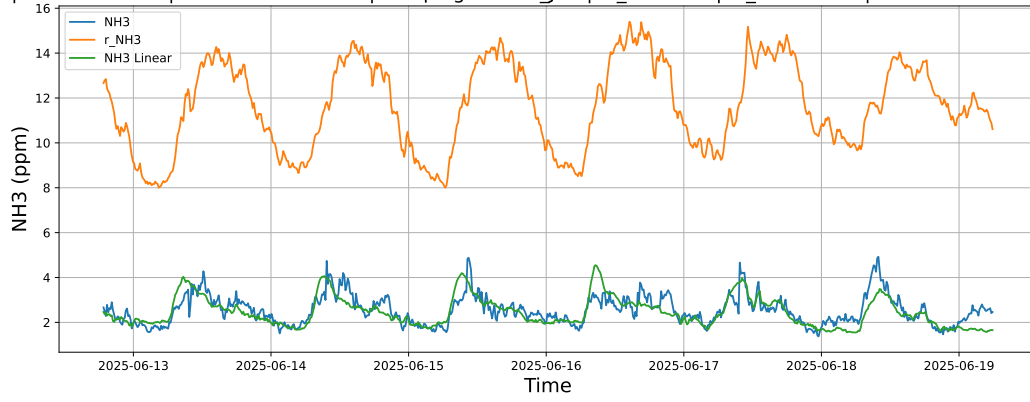


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

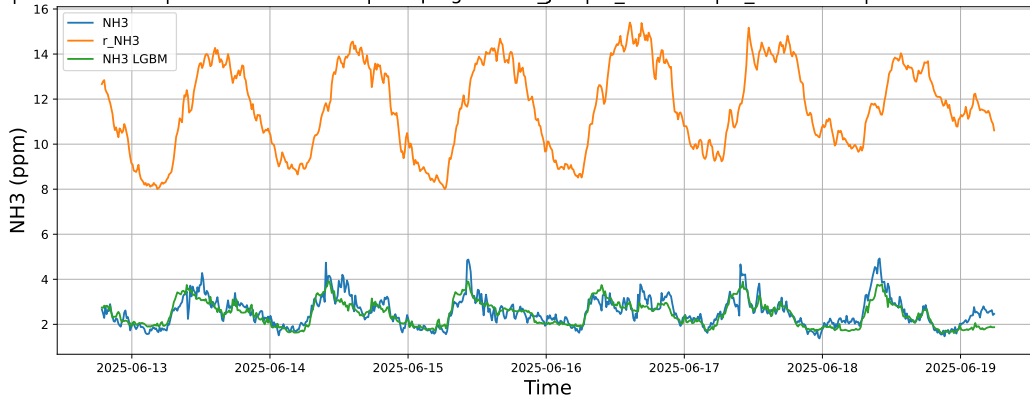


4.1.3.3 Time window : ALL

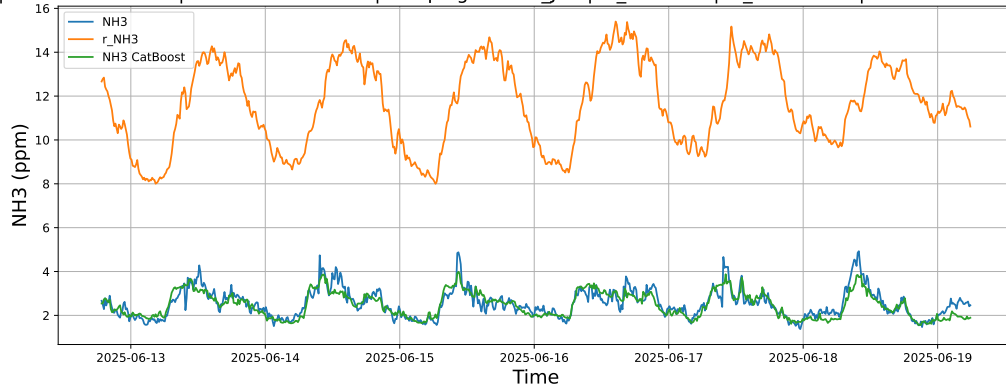
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



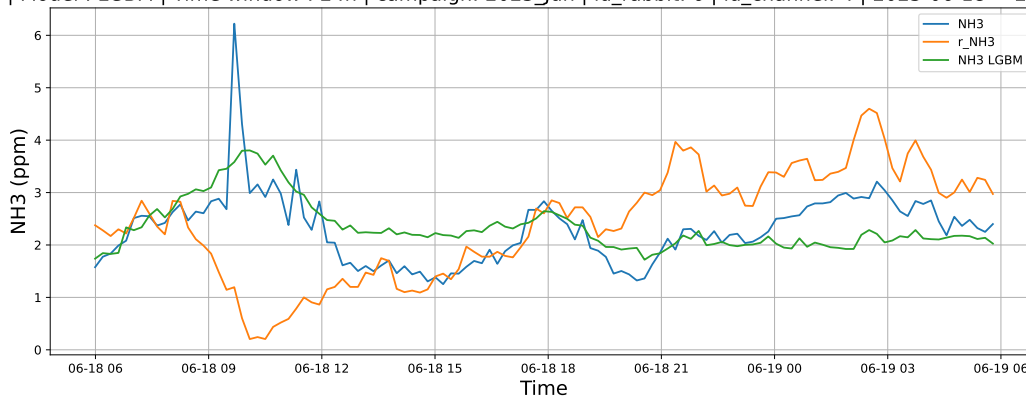
4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

4.1.4.1 Time window : 24

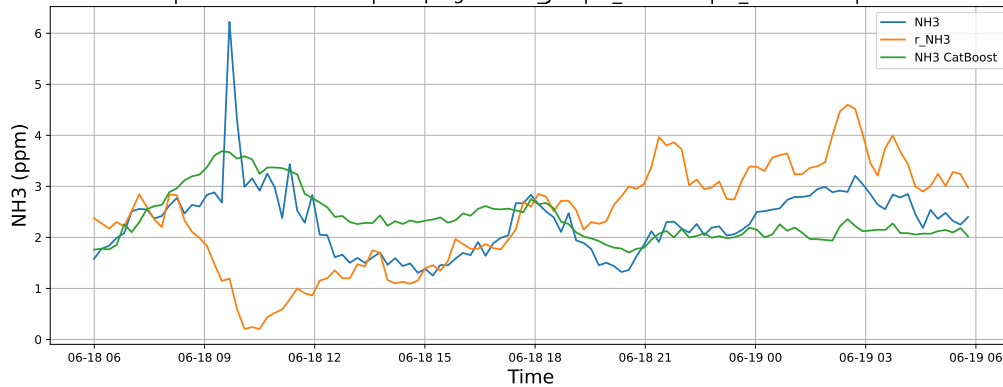
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

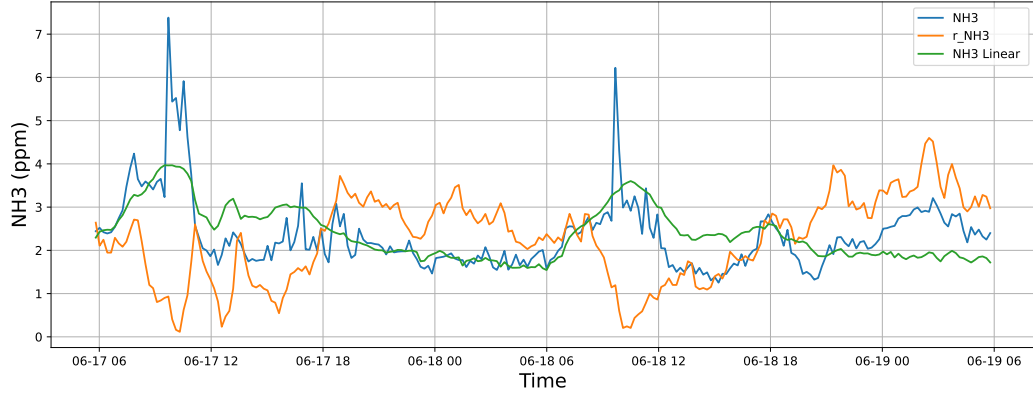


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

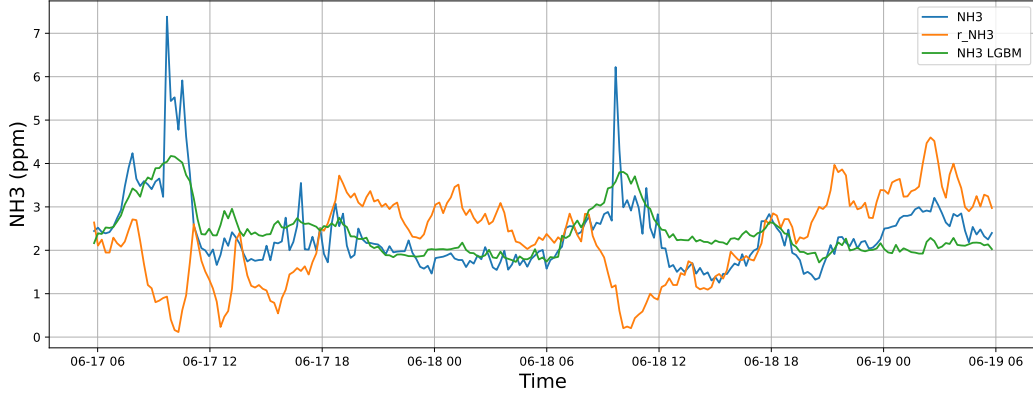


4.1.4.2 Time window : 48

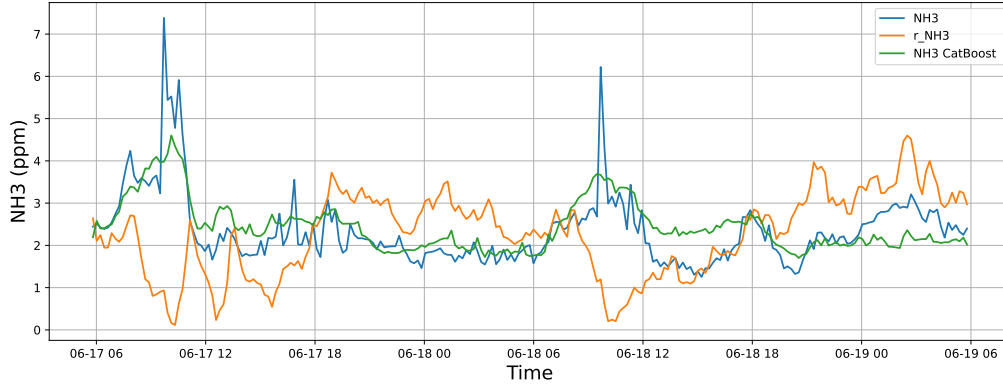
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

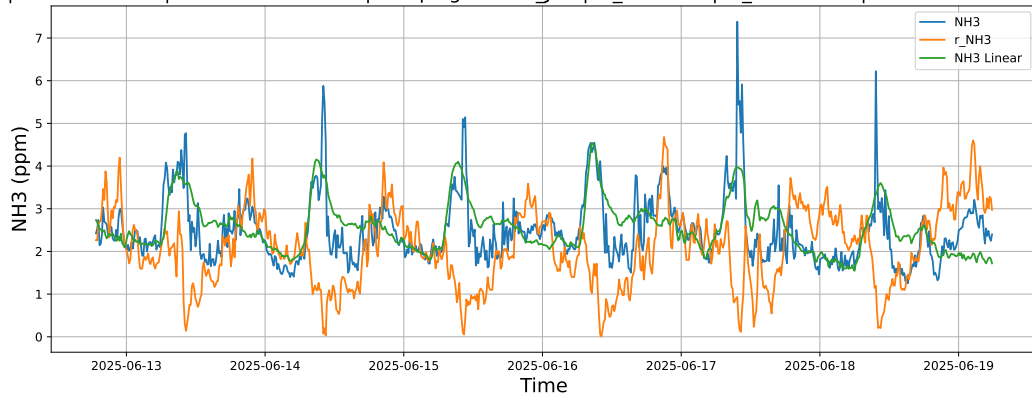


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

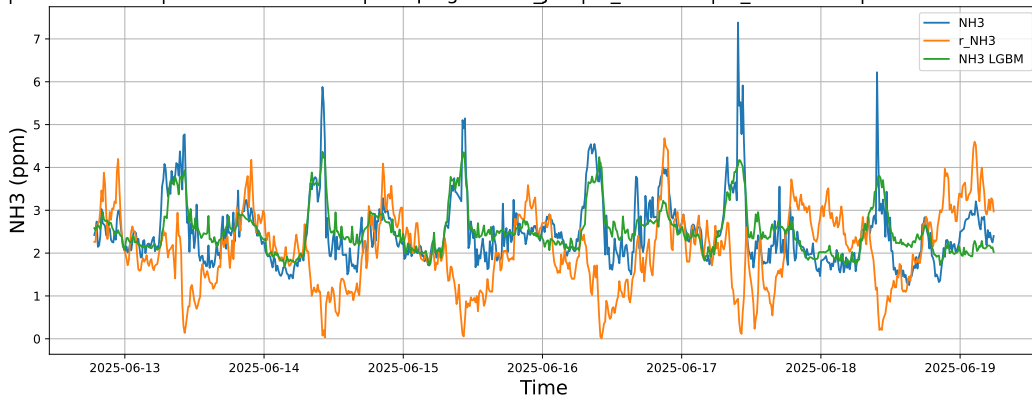


4.1.4.3 Time window : ALL

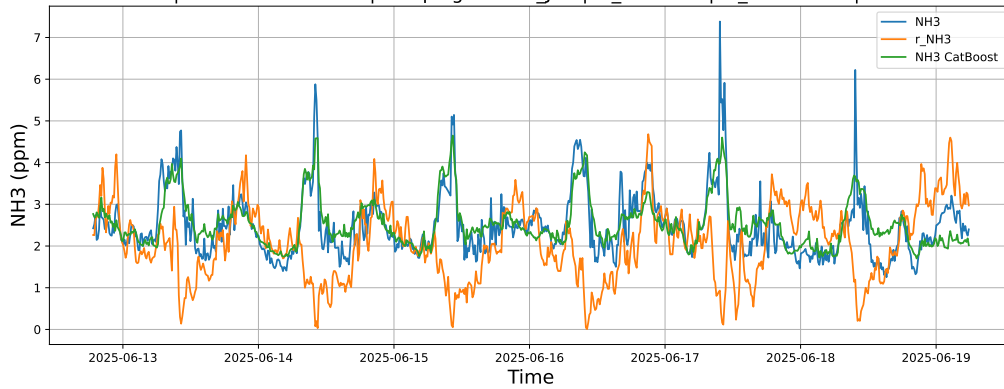
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



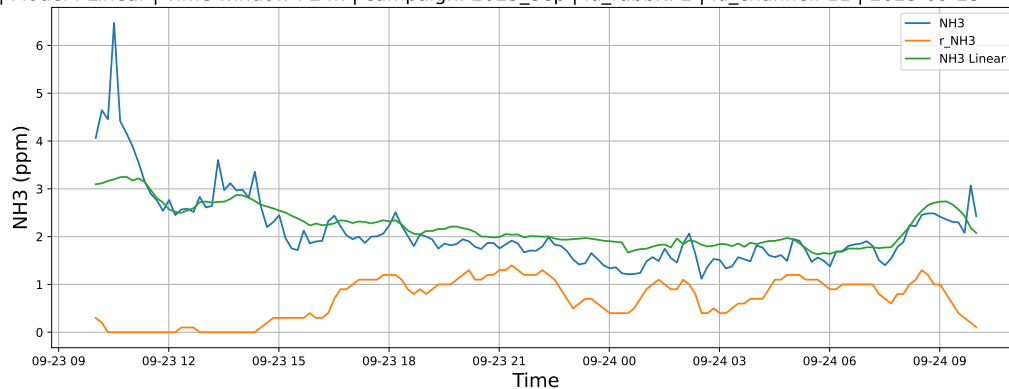
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



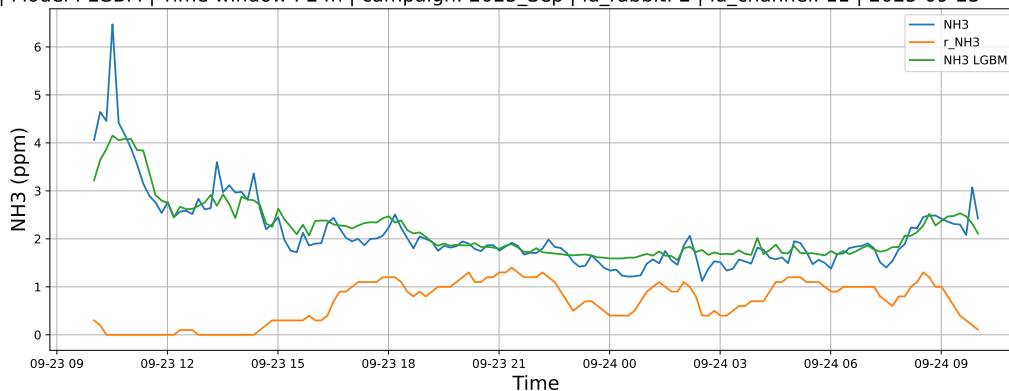
4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

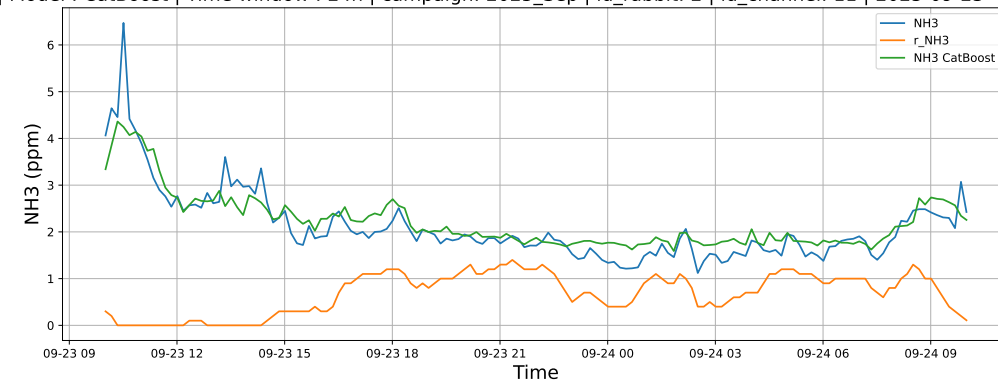
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

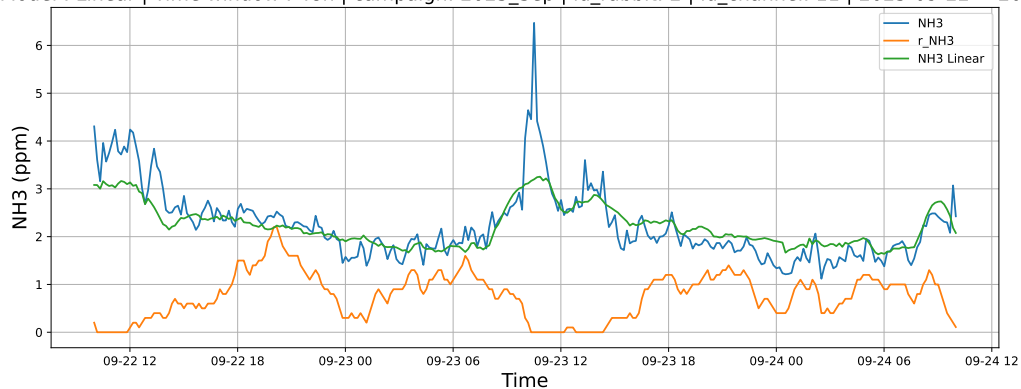


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

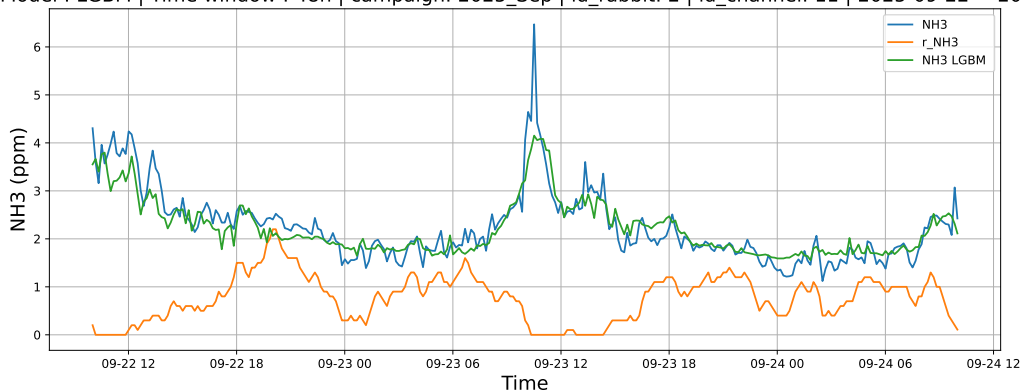


4.1.5.2 Time window : 48

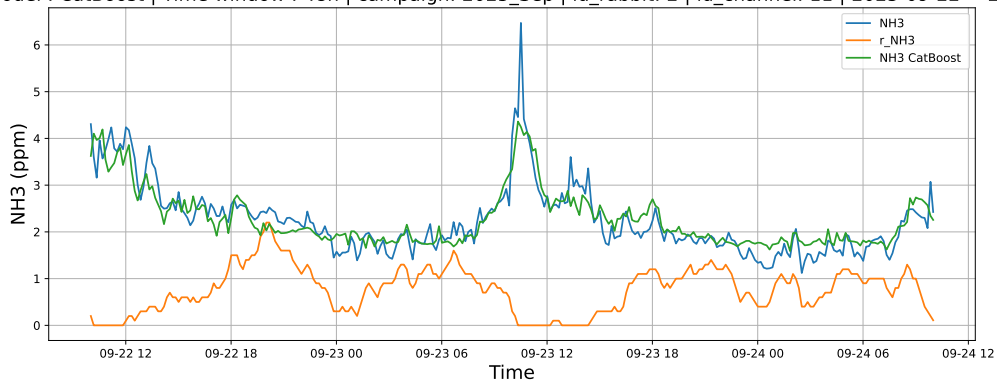
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

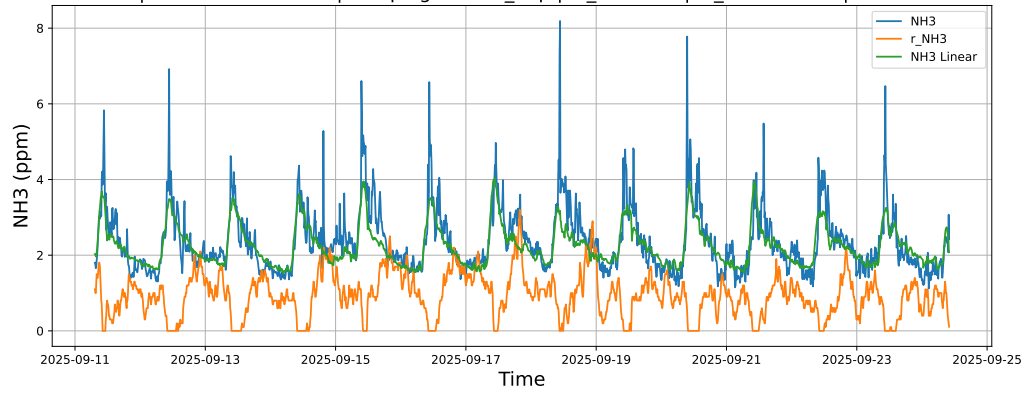


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

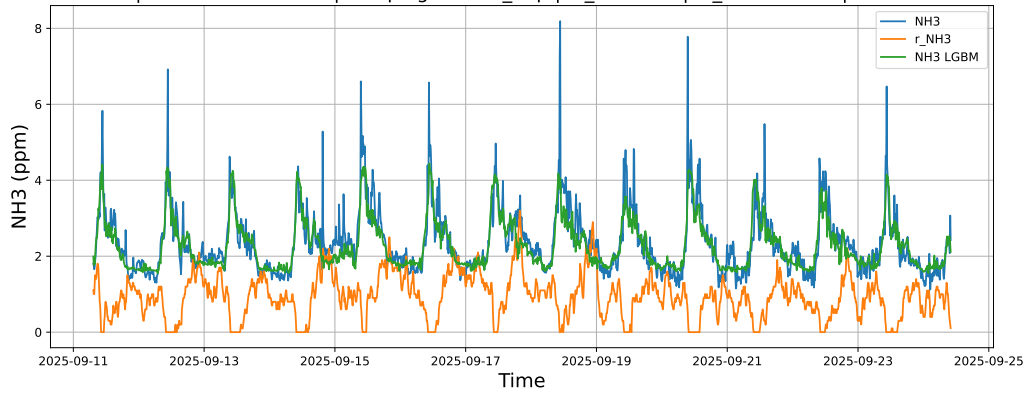


4.1.5.3 Time window : ALL

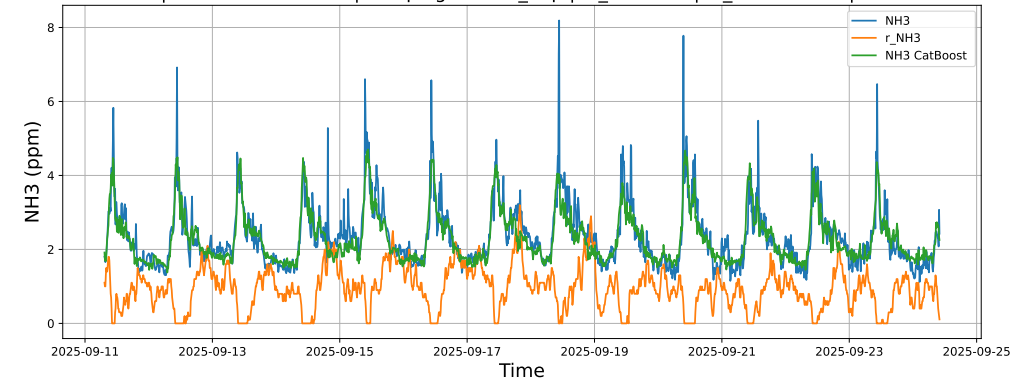
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



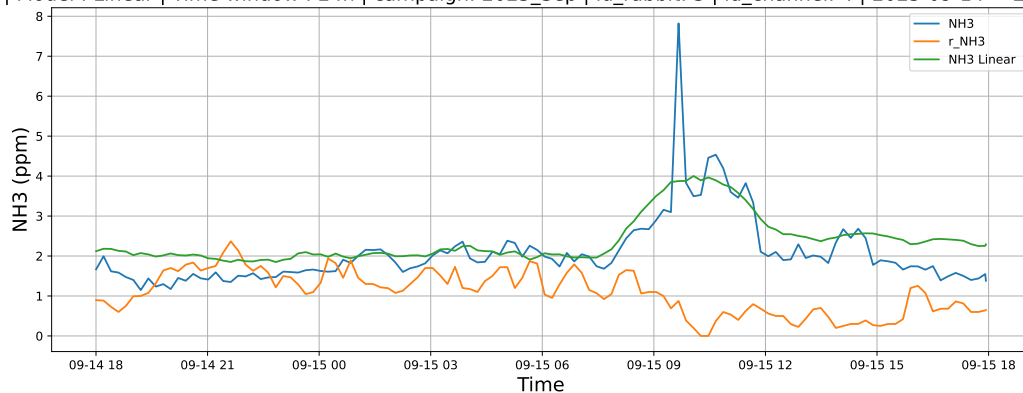
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



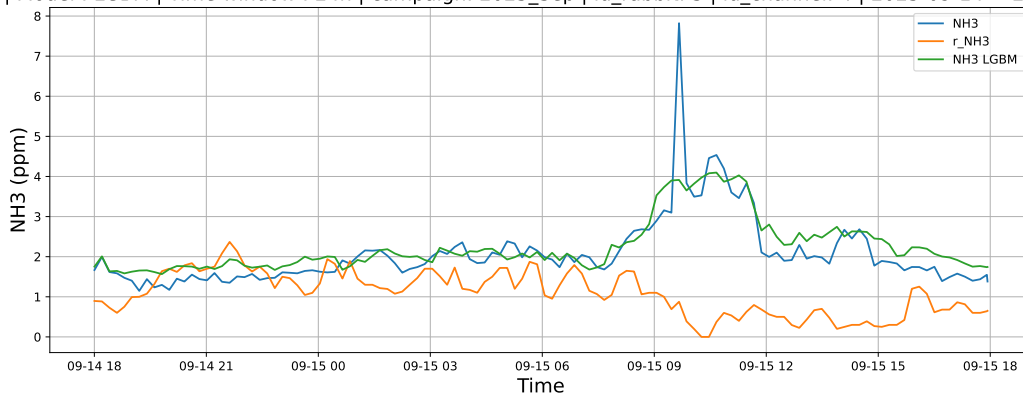
4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.1.6.1 Time window : 24

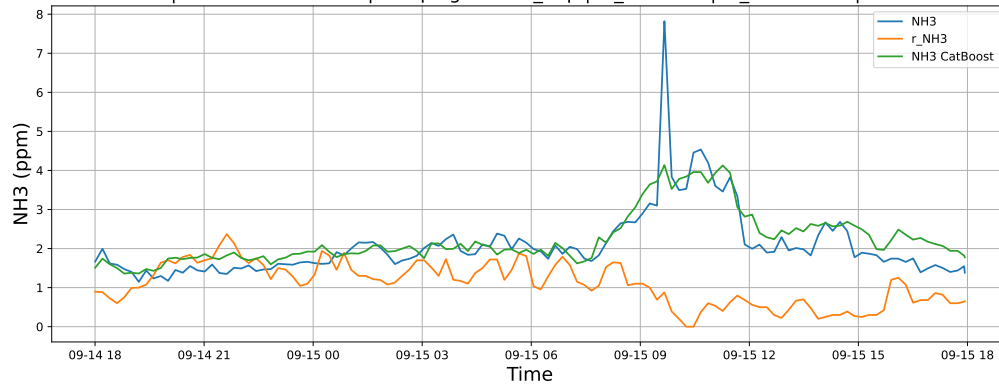
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

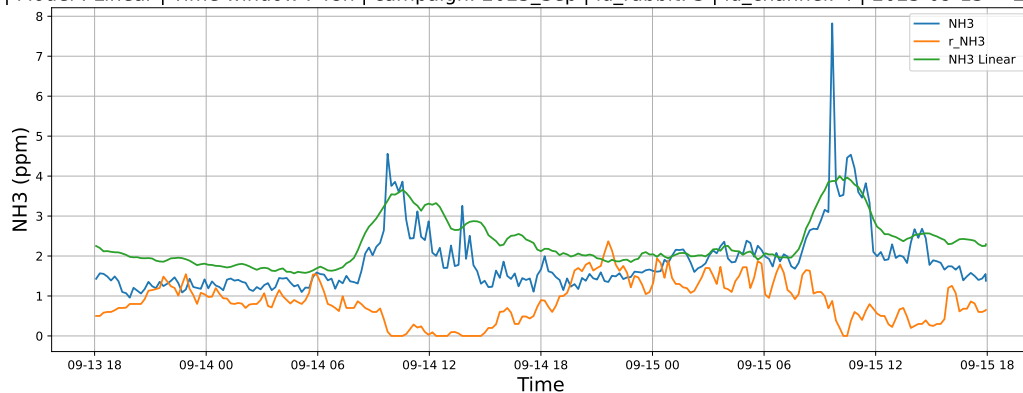


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

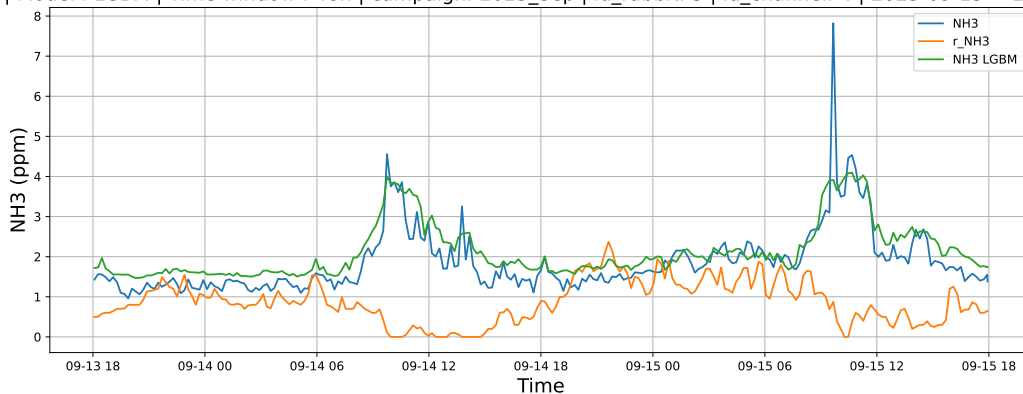


4.1.6.2 Time window : 48

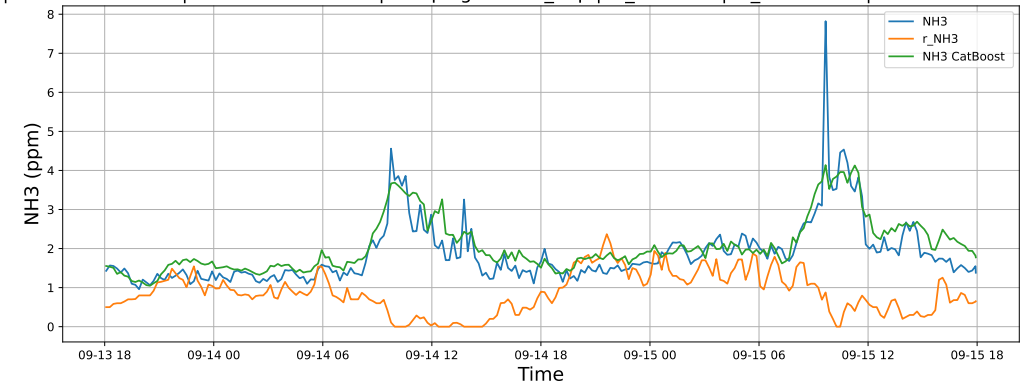
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

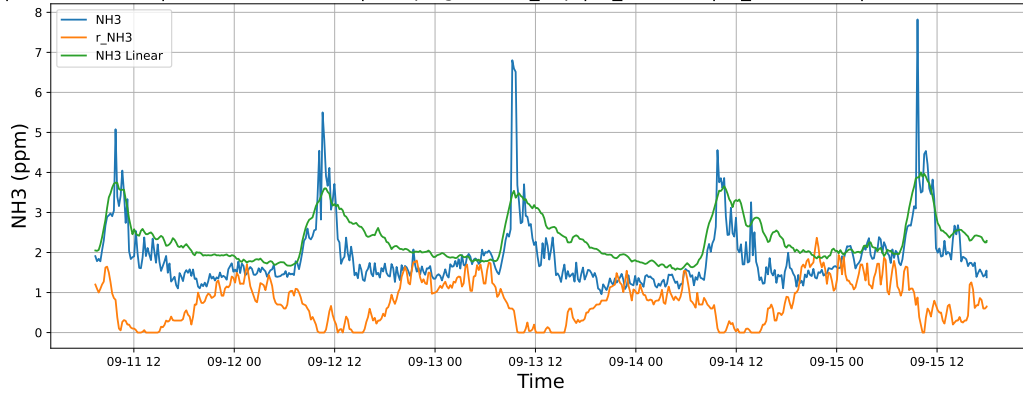


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

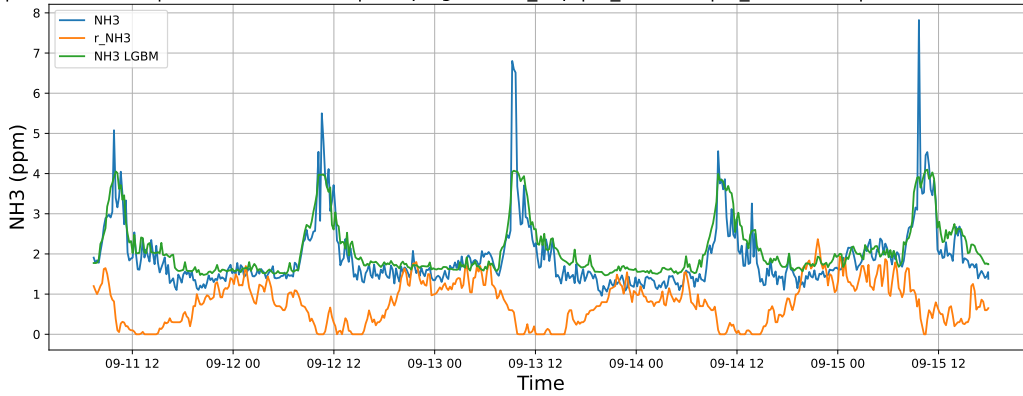


4.1.6.3 Time window : ALL

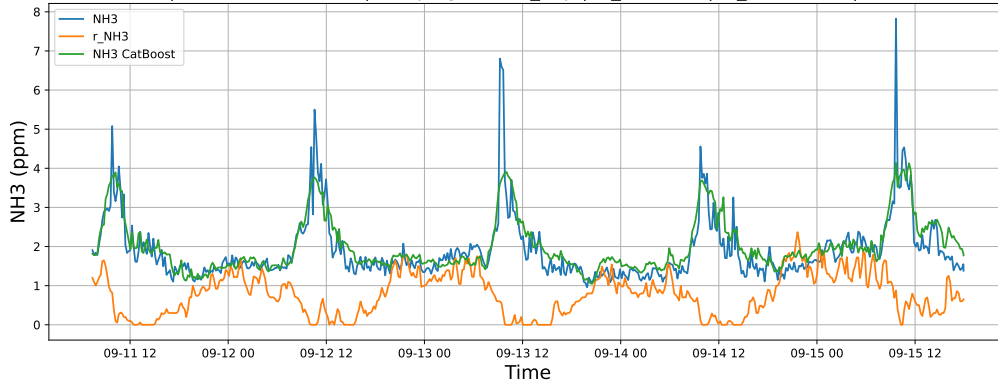
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



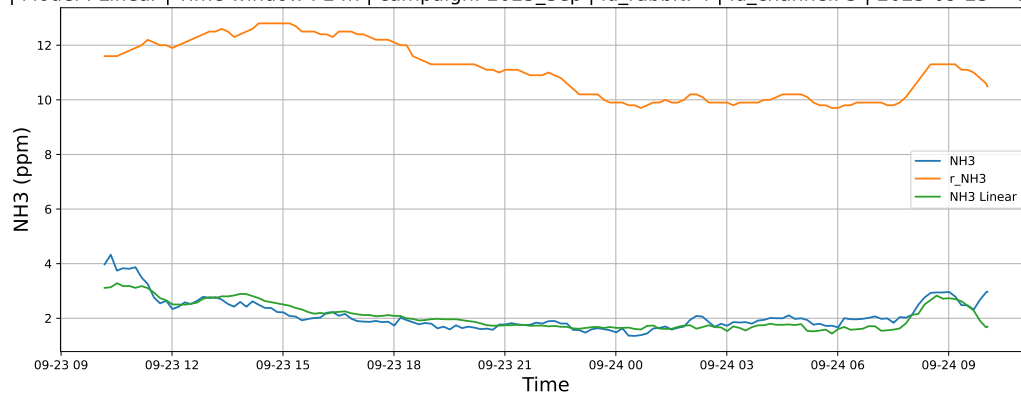
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



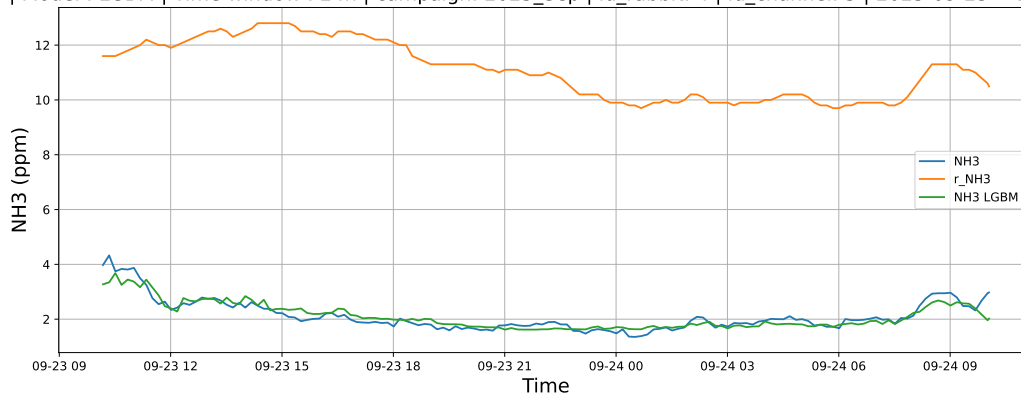
4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

4.1.7.1 Time window : 24

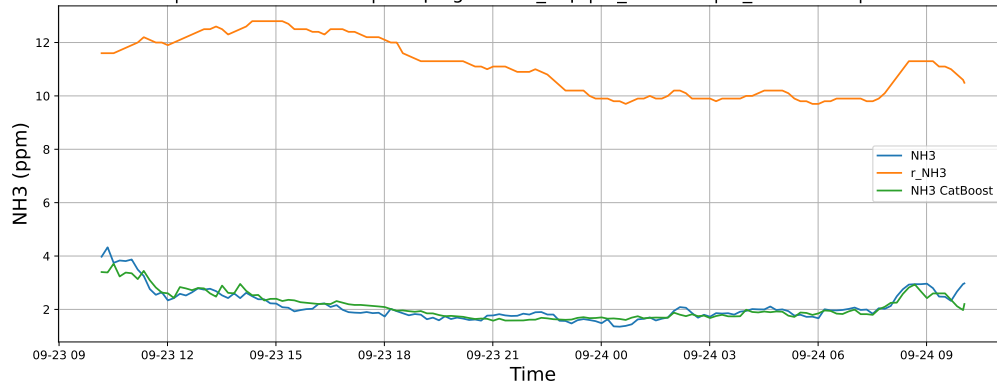
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

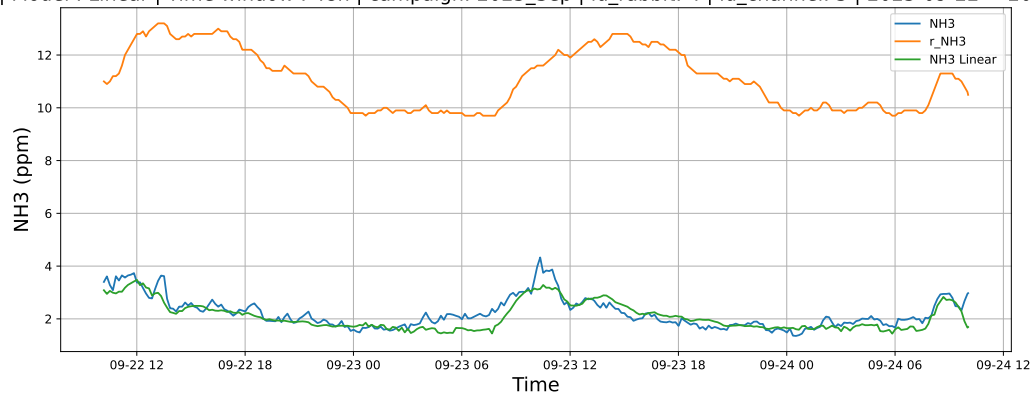


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

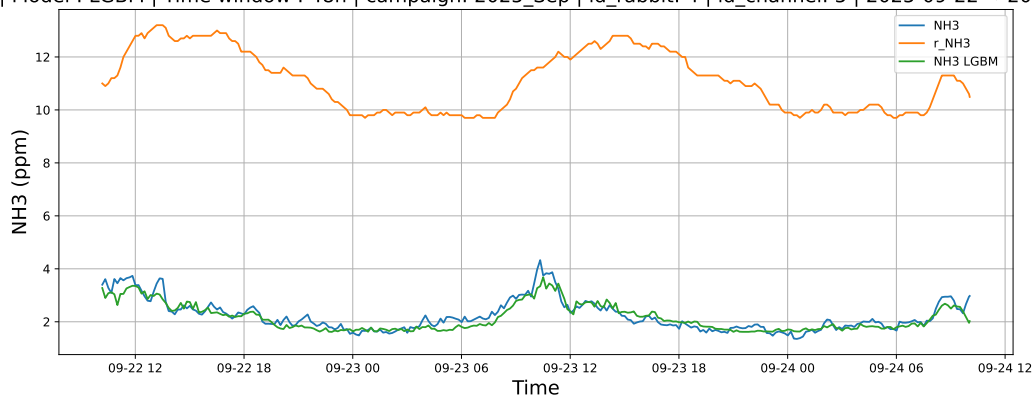


4.1.7.2 Time window : 48

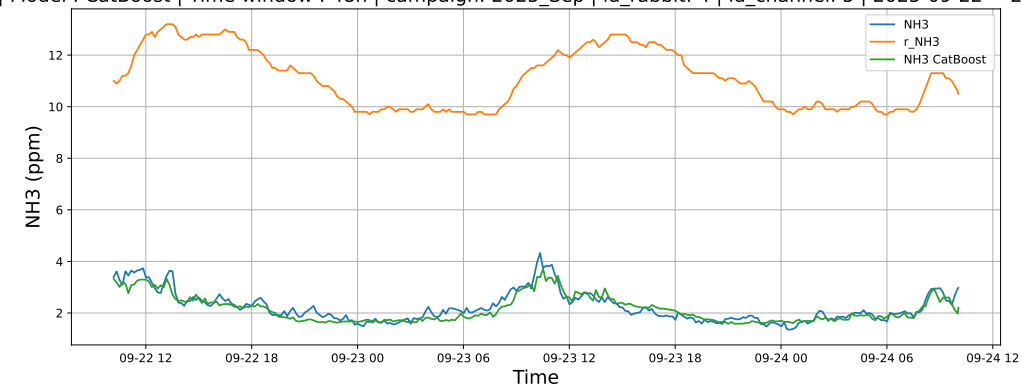
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

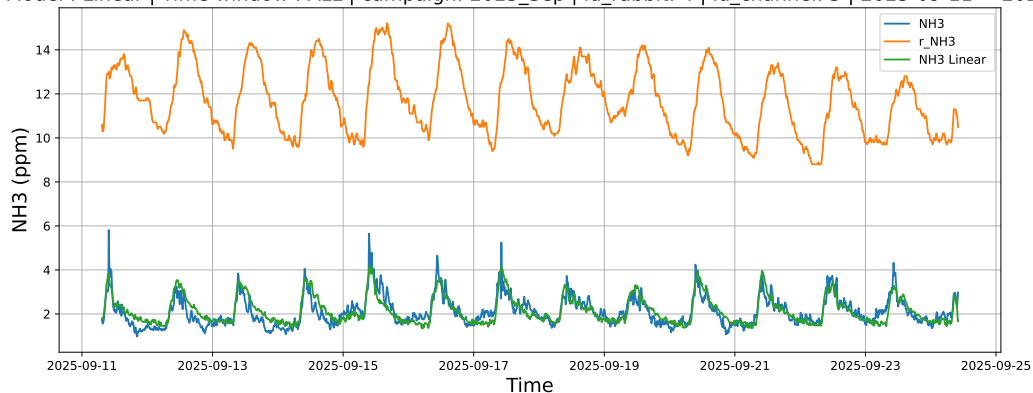


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

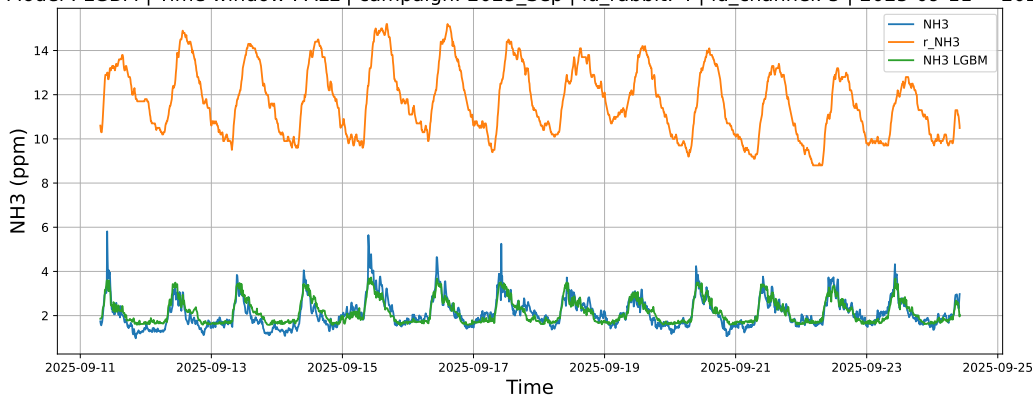


4.1.7.3 Time window : ALL

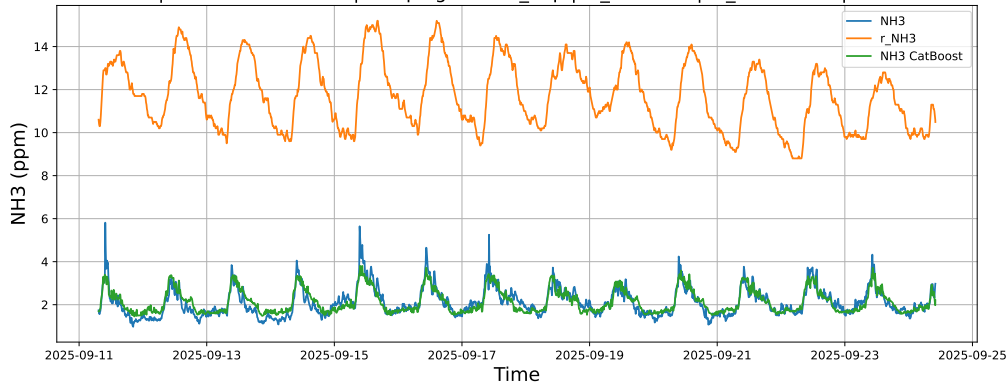
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



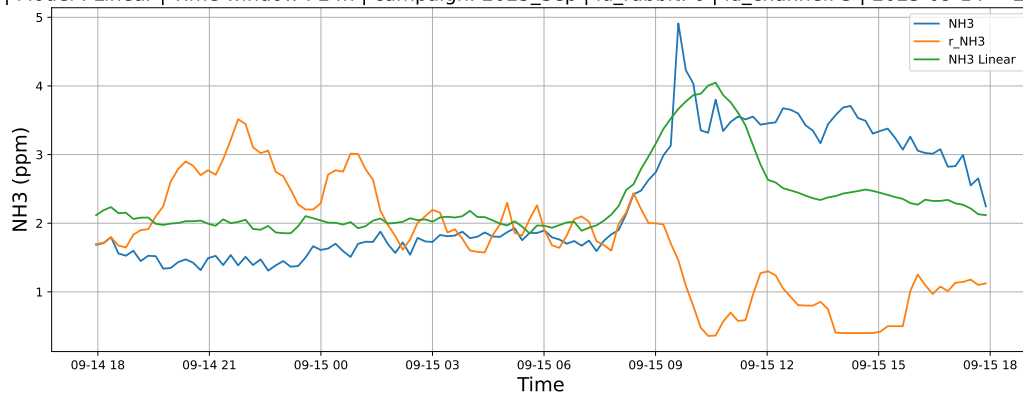
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



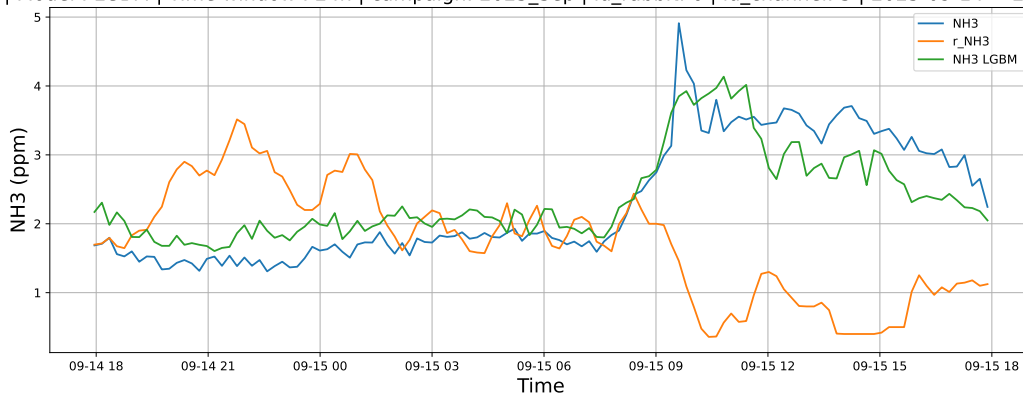
4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

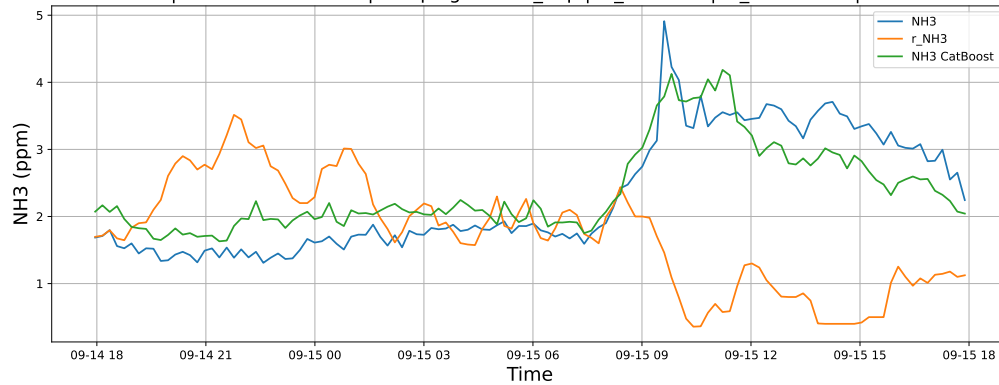
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

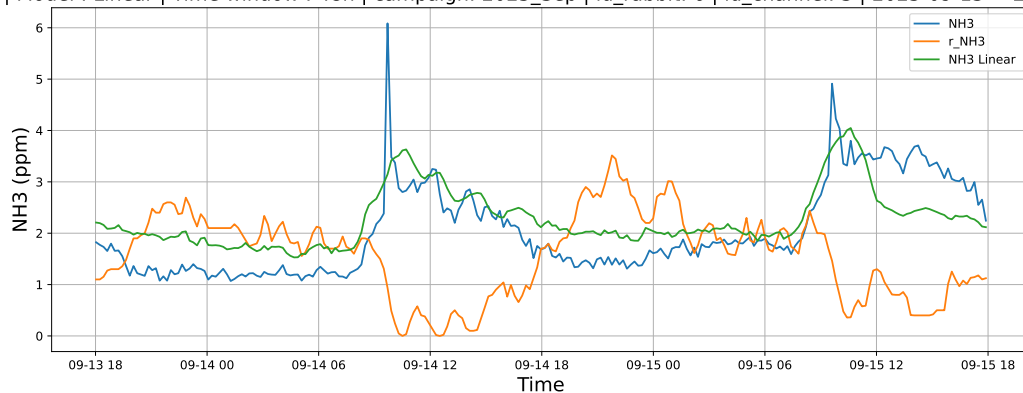


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

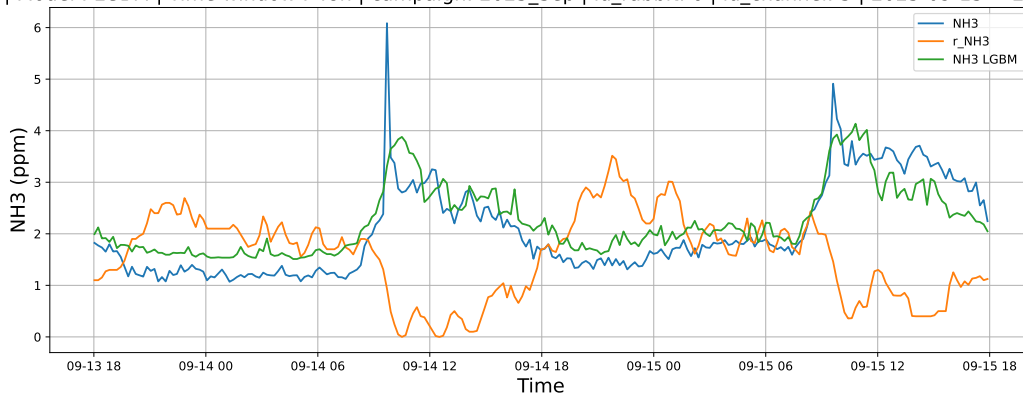


4.1.8.2 Time window : 48

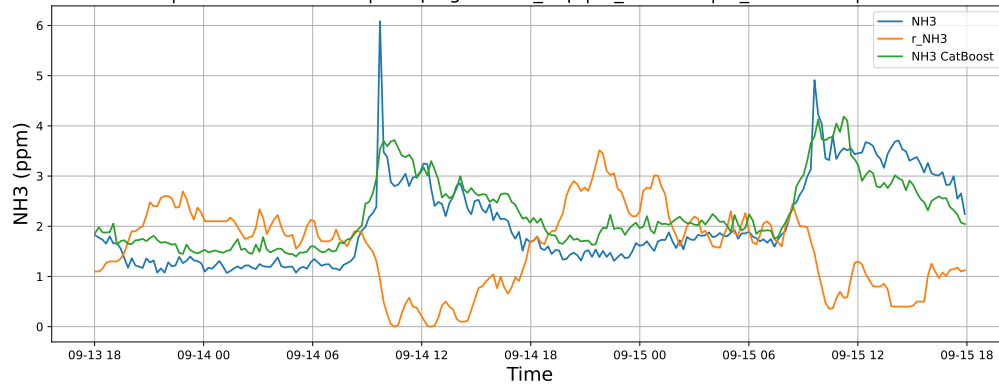
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

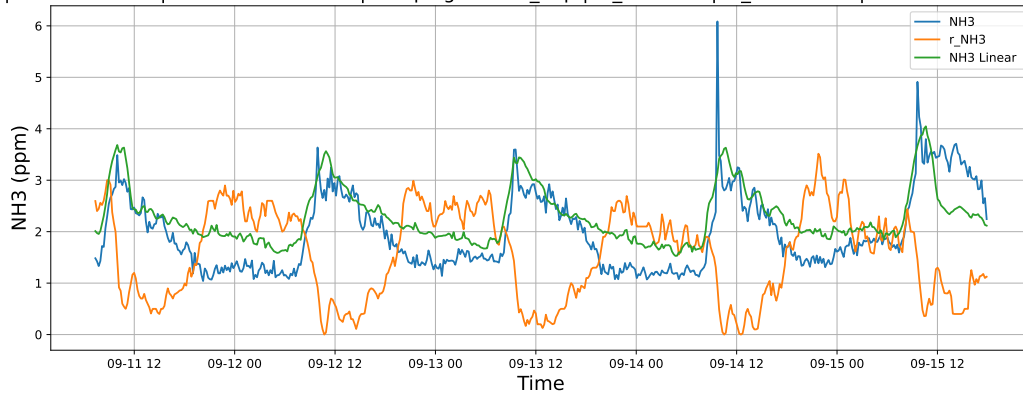


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

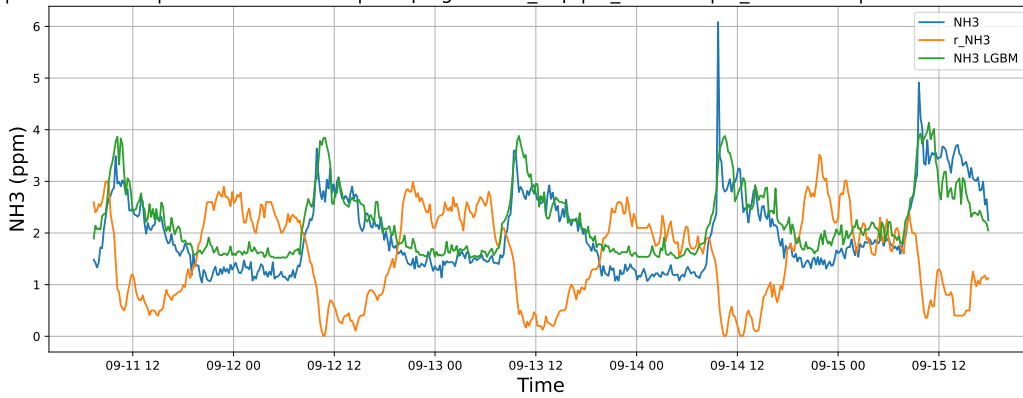


4.1.8.3 Time window : ALL

NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

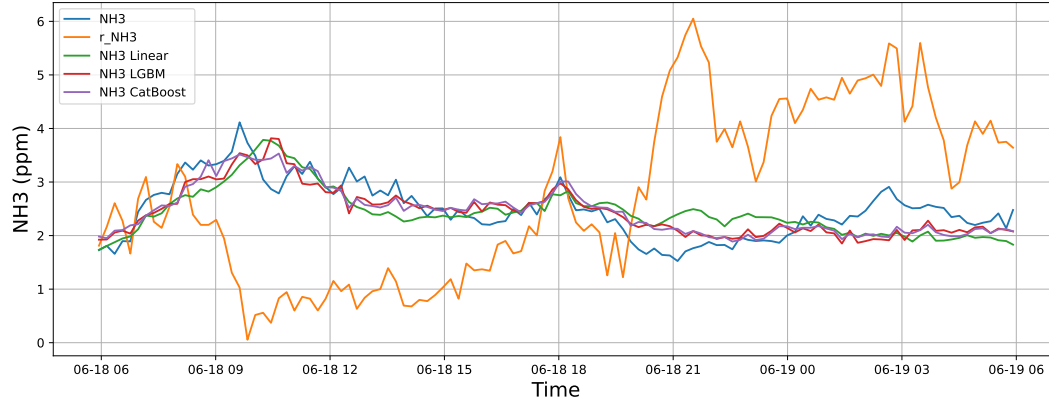


4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

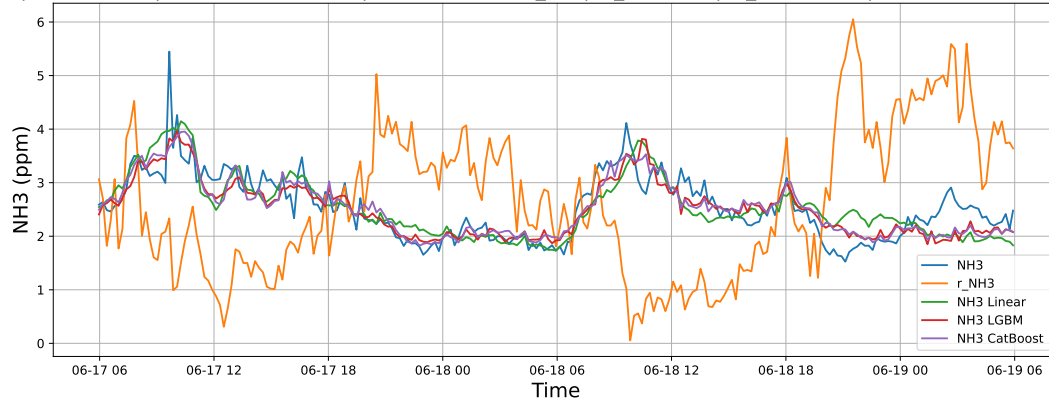
4.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



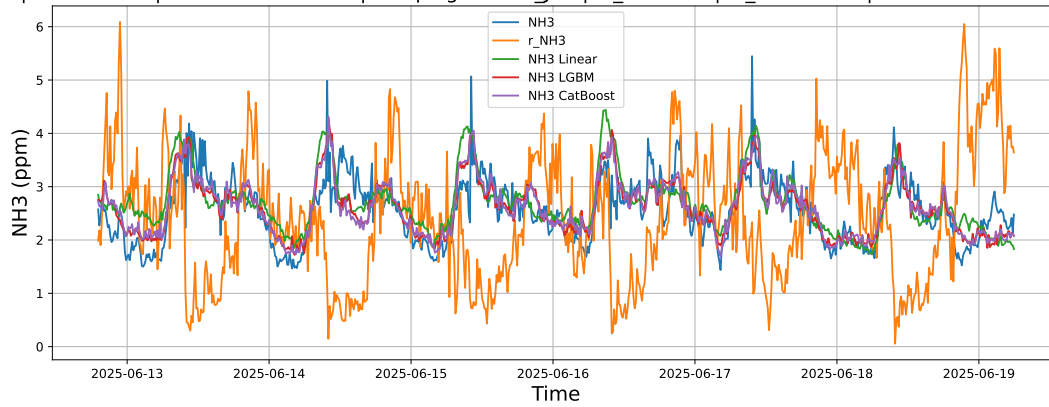
4.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



4.2.1.3 Time window : ALL

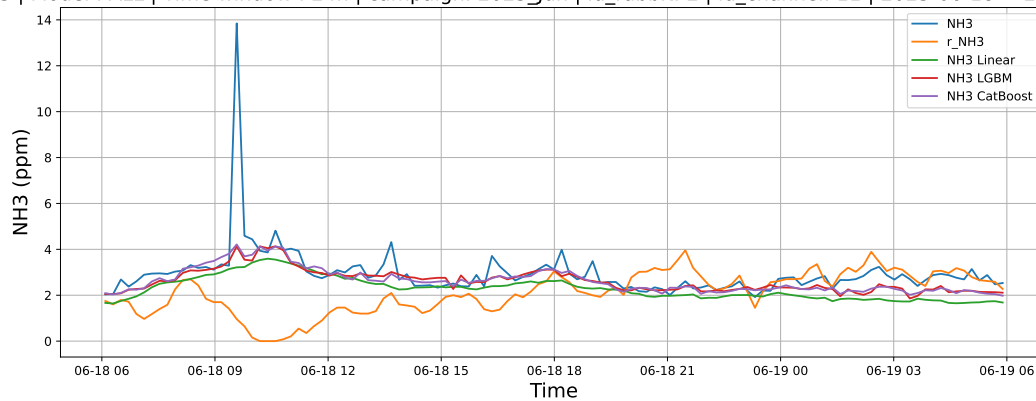
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

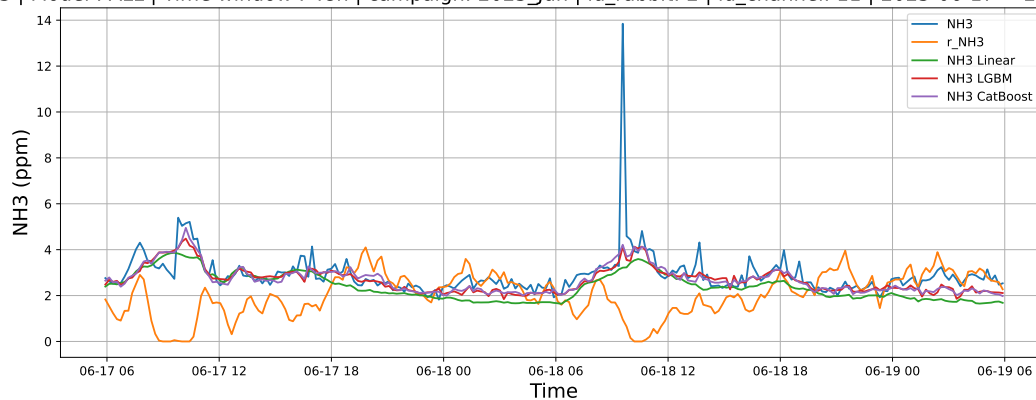
4.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



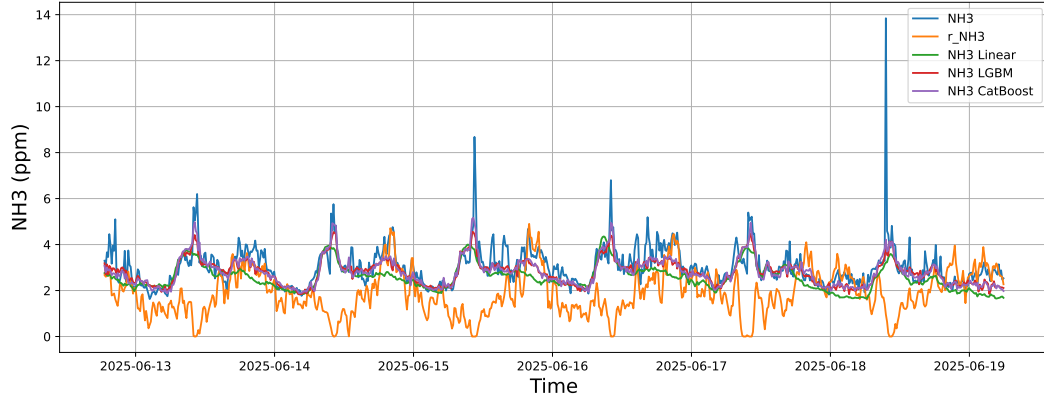
4.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



4.2.2.3 Time window : ALL

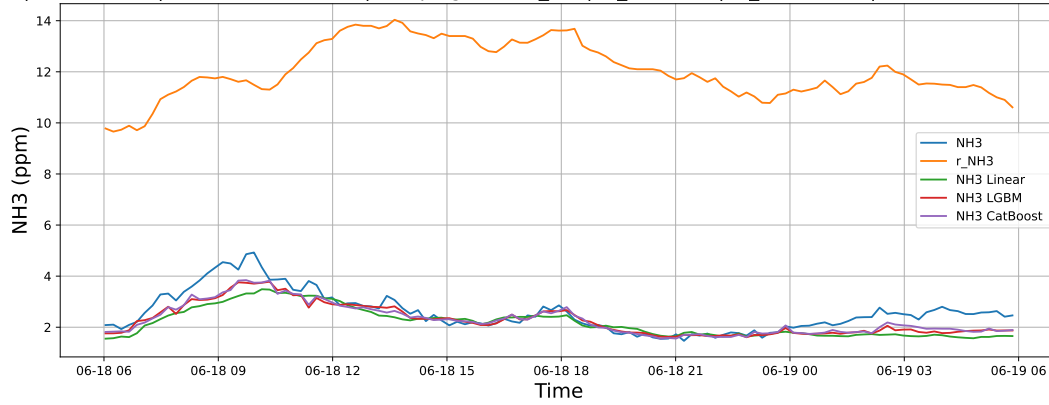
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

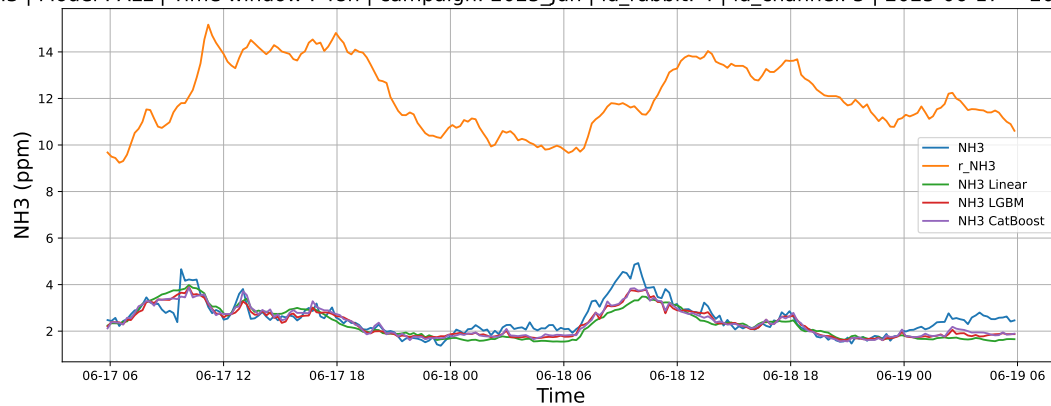
4.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



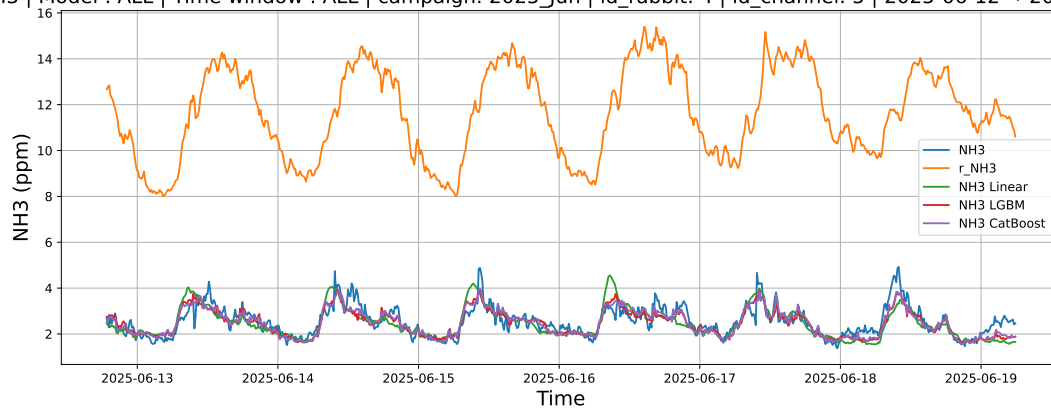
4.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



4.2.3.3 Time window : ALL

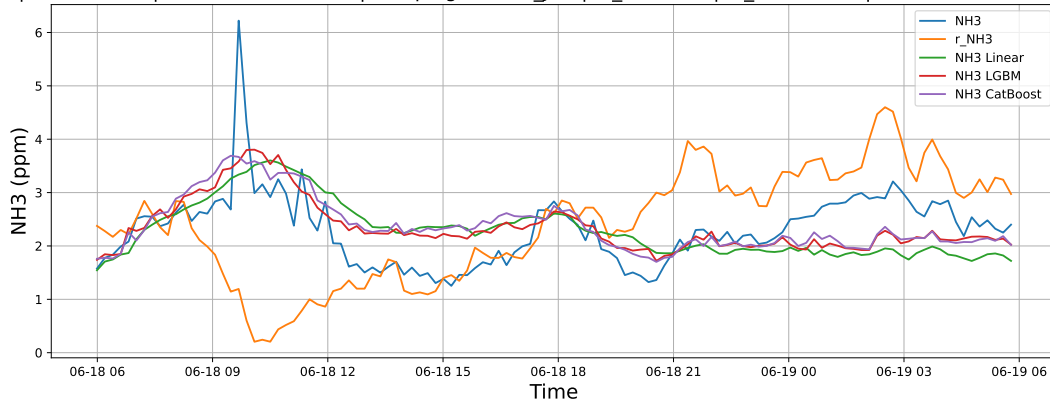
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

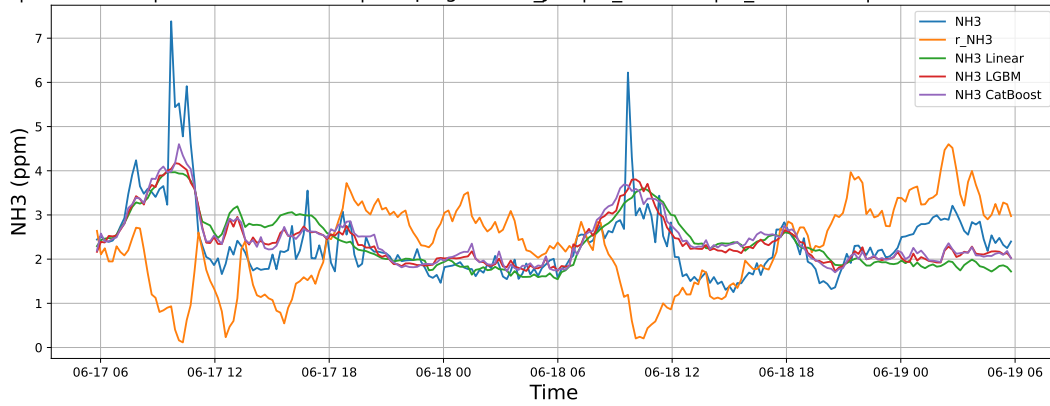
4.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



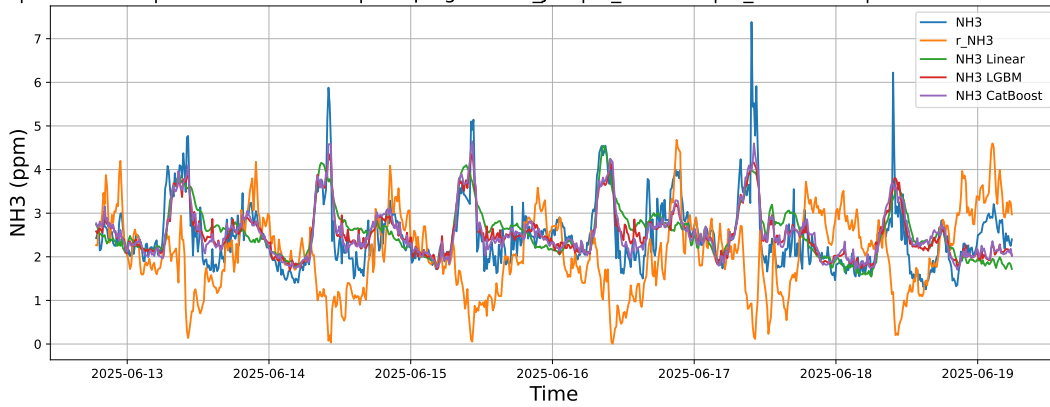
4.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



4.2.4.3 Time window : ALL

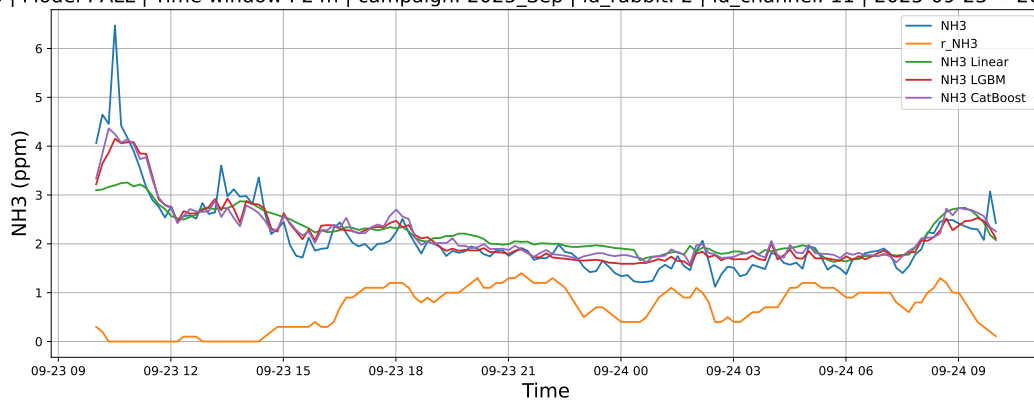
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

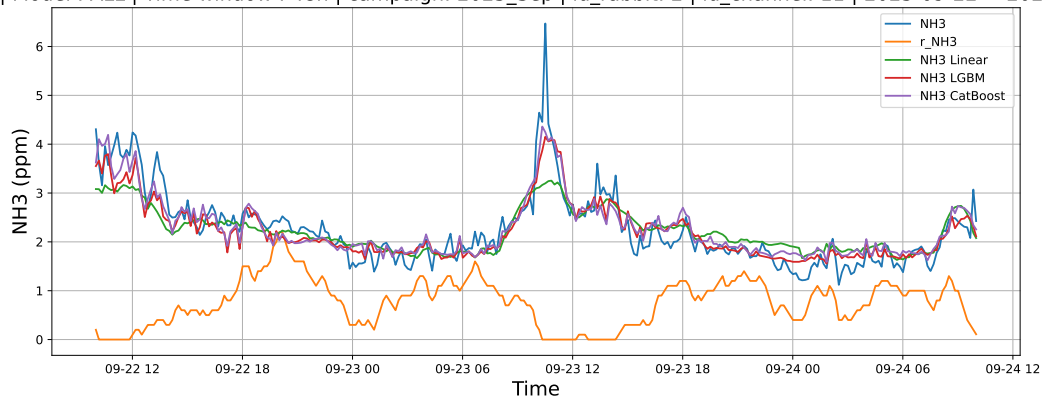
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



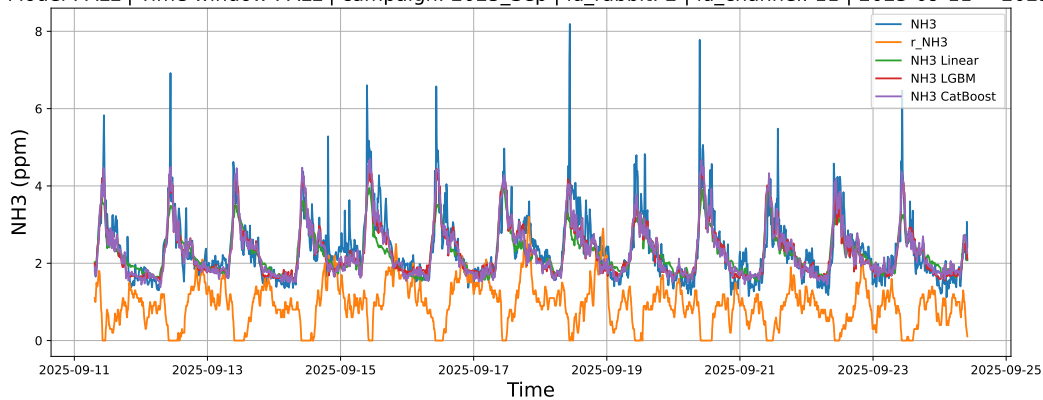
4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



4.2.5.3 Time window : ALL

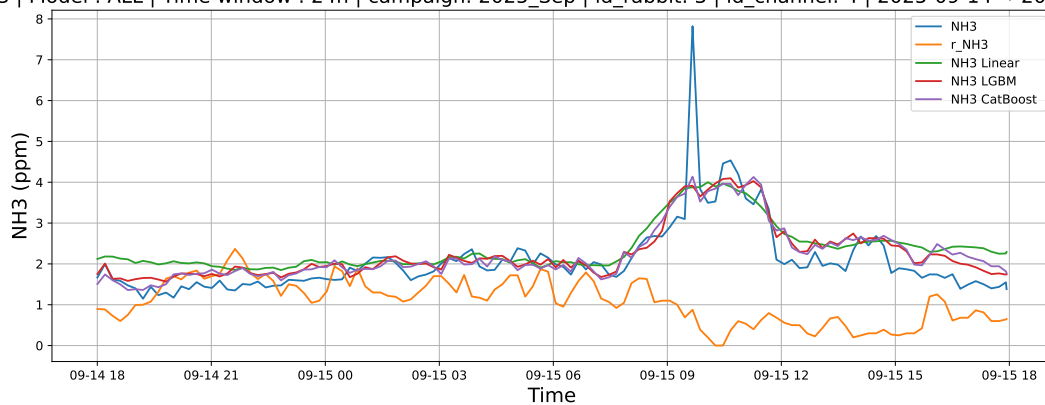
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

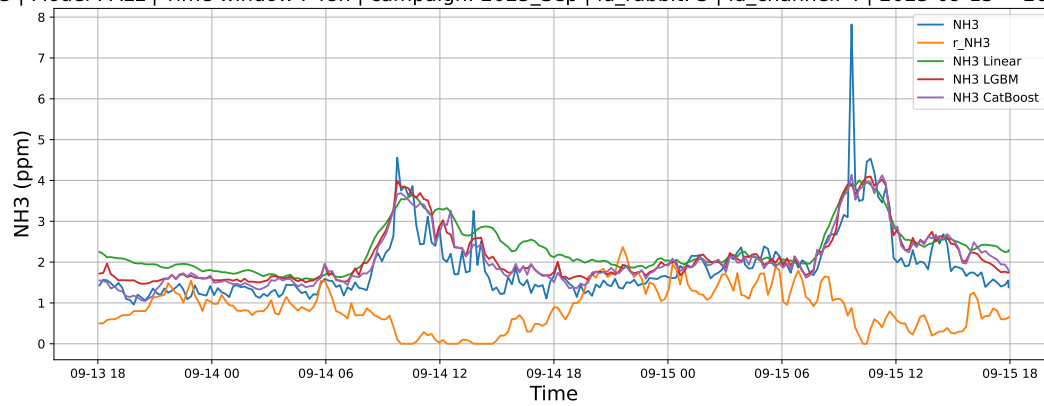
4.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



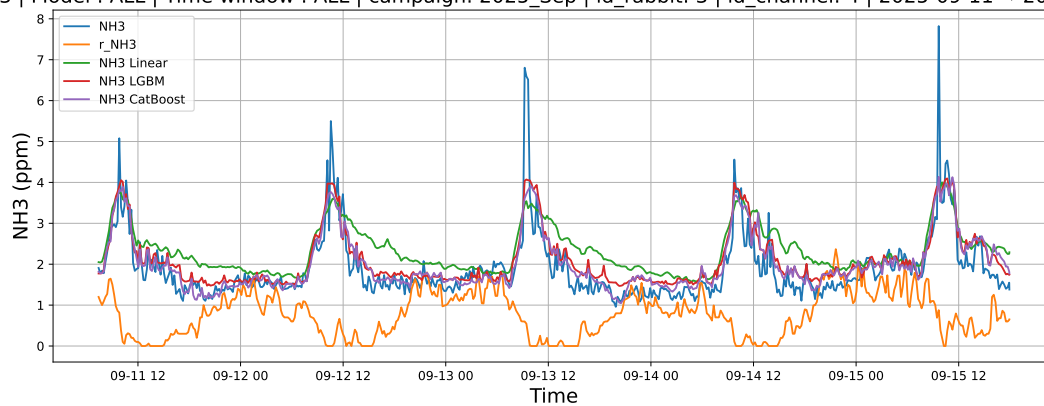
4.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



4.2.6.3 Time window : ALL

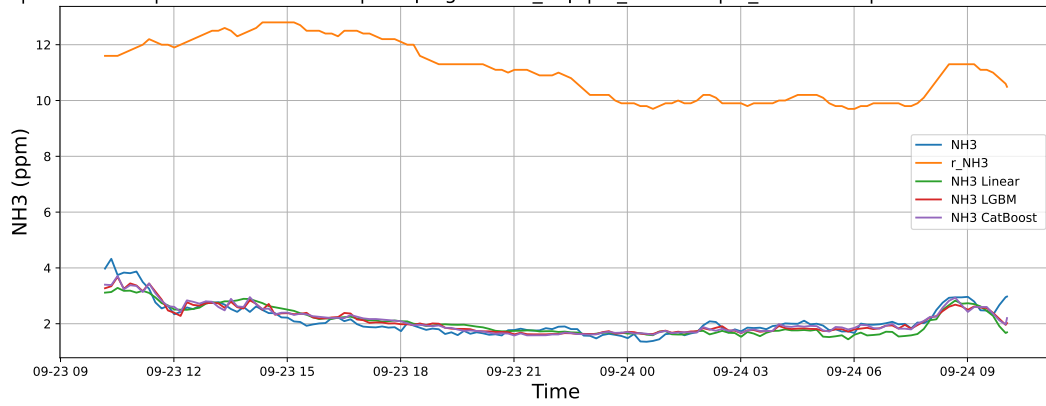
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

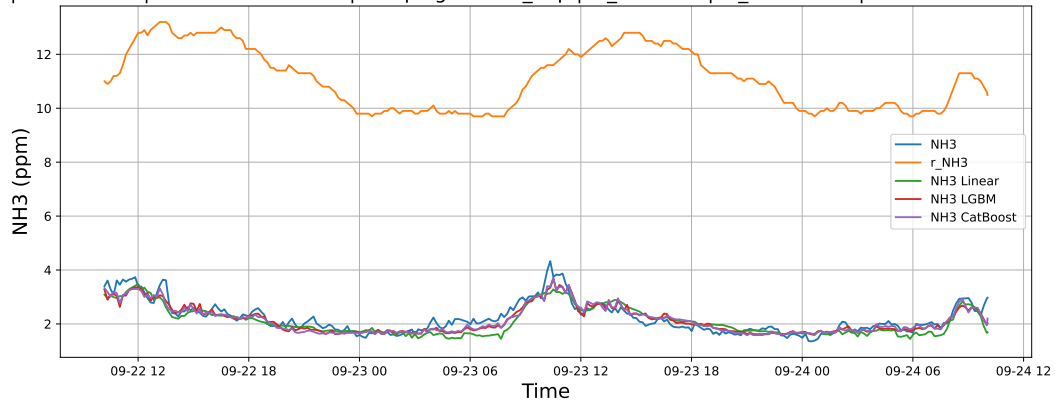
4.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



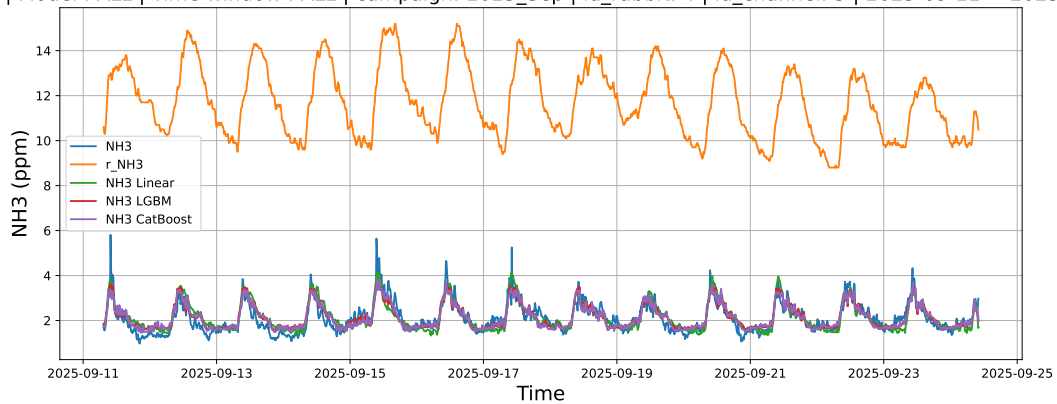
4.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



4.2.7.3 Time window : ALL

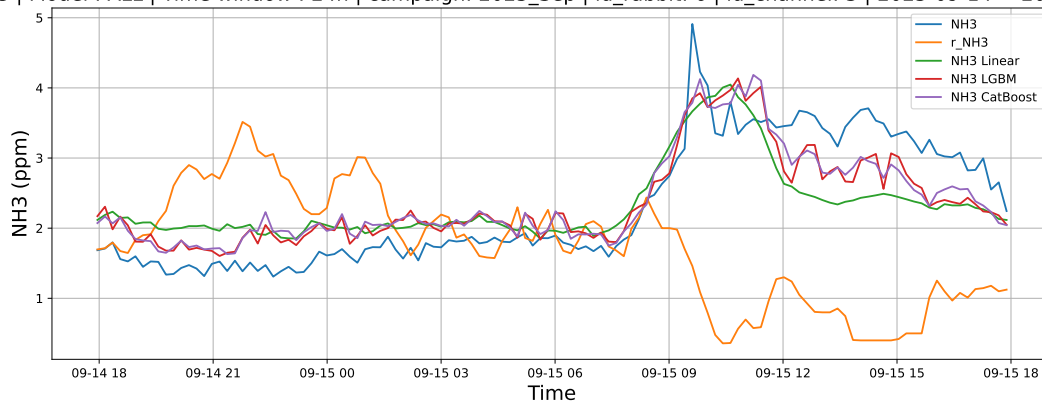
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

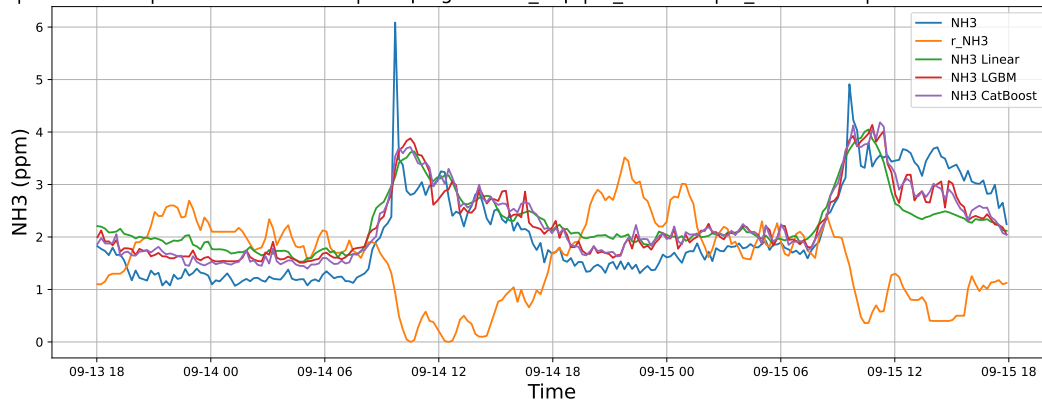
4.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



4.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



4.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

